Supervised Bayesian Matrix Factorization

Joint Gene Expression Programs & Survival Modeling in PDAC

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2026-04-09

Outline

Motivation · The PDAC challenge & why supervised factorisation matters

Methods · SBMF model · Dual-source objective · EBNM adaptive priors · CAVI inference

Results · Synthetic validation · Seven PDAC cohorts · Prior × K comparison · Deep dives

Future Directions & Summary

Overview of today's talk. About 15 minutes of content, then open for discussion. I'll start with why we need supervised approaches for PDAC, walk through the Bayesian model and inference, then present results on 7 independent cohorts.

The PDAC Problem

Pancreatic Ductal Adenocarcinoma

- **Five-year survival $< 12\%$** — one of the most lethal solid tumours
- Most patients diagnosed at advanced stage with limited treatment options
- **No widely adopted molecular stratification** at diagnosis
- Tumours are a complex mixture of malignant, stromal, and immune cells — gene expression signals are deeply interwoven
- **Goal:** identify *gene expression programs* (GEPs) whose activation levels predict patient survival

> **The Clinical Need**
>
> Reliable prognostic signatures could:
>
> - Stratify patients for targeted therapy
> - Define subgroups for clinical trial enrollment
> - Distinguish molecular features that *describe* tumour heterogeneity from those that *predict* disease course

PDAC is the most lethal major solid tumour. Unlike breast cancer where molecular subtypes dictate treatment, PDAC patients largely receive the same regimen. If we can identify gene programs that distinguish prognostic subgroups, we can stratify patients for targeted therapy and clinical trials. The complexity is that bulk expression mixes signals from tumour, stroma, and immune cells.

Why Supervised > Unsupervised

The PCA-Then-Test Gap

Standard approach:

1. Apply PCA or NMF to expression data
2. Find axes of greatest expression variance
3. Test for survival association *post hoc*

The problem:

- Expression-dominant axes may reflect batch effects or housekeeping genes
- Programs with **modest expression variance** but **strong survival signal** get missed
- Post-hoc testing is statistically conservative

Supervised approach:

1. Integrate survival into the factorisation objective *directly*
2. Factors are simultaneously constrained to explain expression **and** predict survival
3. If $\hat{\beta}_k = 0$, factor k explains expression only — the model naturally separates prognostic from non-prognostic structure

> Key Insight
>
> A factor explaining 15% of variance might be entirely batch effects. A factor explaining 3% might perfectly separate survivors from non-survivors. PCA prioritises the first and ignores the second.

PCA finds axes of greatest expression variance, not greatest clinical relevance. A survival-supervised approach forces factors to be useful for prediction. If a factor carries expression signal but no survival information, the model shrinks its beta to zero — we get the separation for free.

The Gap & Our Approach

What Existing Supervised MF Approaches Are Missing

- **Data-adaptive shrinkage priors** — existing methods use fixed penalties; we let the prior learn from the data
- **Automatic K selection** — no need to grid-search the number of factors
- **Uncertainty quantification** — posterior inference gives us credible intervals, not just point estimates

SBMF vs. Penalised Supervised Approaches

Penalised approaches (e.g., deSurv from our lab) optimise a weighted objective $(1-\alpha)\mathcal{L}_{\text{NMF}} + \alpha\mathcal{L}_{\text{Cox}}$, requiring a tuning parameter α to balance the two terms. **SBMF** instead uses full Bayesian inference:

- The **ELBO** naturally balances genomics and survival — no α needed
- **EBNM** priors adapt their shrinkage to the data
- Posterior second moments ($E[L^2] \neq \bar{L}^2$) prevent overfitting to uncertain loadings

SBMF: Learn gene expression programs and survival associations simultaneously, with the prior learning from the data.

This is the only slide where I reference deSurv — Amber's work from our lab. The key distinction is that SBMF is fully Bayesian: we use the ELBO which naturally balances both objectives without a tuning alpha, and EBNM priors that adapt their shrinkage to the data. The posterior second moments are critical for preventing overfitting when loadings are uncertain.

The SBMF Model

Joint Genomics + Survival Factorisation

$$Y = LF^\top + E, \quad E_{ij} \sim N(0, \tau_j^{-1})$$

$$h_i(t) = h_0(t)\exp(L_i \cdot \beta)$$

Quantity	Dimension	Meaning
Y	$n \times p$	Gene expression matrix
L	$n \times K$	Patient activation scores
F	$p \times K$	Gene weights per program
β	$K \times 1$	Survival coefficients
τ_j	scalar	Gene-specific precision

The Shared Latent Space

The **same** L drives both:

- Low-rank reconstruction of gene expression ($LF^\top$)
- Cox proportional hazards risk score ($L\beta$)

This forces factors to explain expression variance **and** maximise survival signal simultaneously.

The model has two components sharing the latent loadings L. The first equation is standard matrix factorisation — Y is approximated by L times F-transpose plus gene-specific noise. The second is a Cox proportional hazards model where the same patient loadings L, multiplied by survival coefficients beta, give the log hazard ratio. Each column of F defines a gene expression program. Each beta_k says whether that program is prognostically relevant.

Dual-Source Objective Function

How the Joint Objective Is Constructed

Genomics Likelihood

$$\mathcal{L}_{\text{genomics}} = \sum_{j=1}^{p} \log p(Y_{\cdot j} \mid L, F_j, \tau_j)$$

Gaussian reconstruction error — how well $LF^\top$ approximates Y

Survival Likelihood

$$\mathcal{L}_{\text{Cox}} = \sum_{i:\delta_i=1} \left[L_i\beta - \log \sum_{j\in\mathcal{R}_i} \exp(L_j\beta) \right]$$

Cox partial log-likelihood — how well $L\beta$ predicts survival

Combined ELBO

$$\text{ELBO} = \underbrace{E_q[\mathcal{L}_{\text{genomics}}]}_{\text{expression fit}} + \underbrace{E_q[\mathcal{L}_{\text{Cox}}]}_{\text{survival fit}} - \underbrace{D_{\text{KL}}(q\|\pi)}_{\text{prior regularisation}}$$

The ELBO **naturally balances** both objectives — no tuning parameter α required. The shared L appears in both terms.

The ELBO is the variational objective we maximise. It decomposes into three intuitive pieces: how well we reconstruct expression, how well we predict survival, and a KL penalty that keeps posteriors close to the prior. The shared L in both likelihood terms is what makes this supervised — the loadings must serve both masters simultaneously.

EBNM & Adaptive Shrinkage

From Penalised Regression to Empirical Bayes

What you already know:

- **Ridge/LASSO** shrink coefficients toward zero with a *fixed* penalty λ
- λ controls how aggressively we regularise
- Chosen by cross-validation — one λ for all coefficients

The empirical Bayes step:

- What if the prior *learned from the data* how much to shrink?

- **EBNM** (Empirical Bayes Normal Means): given noisy observations $x_j = \theta_j + \epsilon_j$, estimate the prior $g(\cdot)$ from the data, then compute posterior $E[\theta_j \mid x_j, \hat{g}]$
- The shrinkage adapts *per factor* and *per gene*

Two prior families tested:

- **Point-normal:** spike at zero + Gaussian slab (soft thresholding, similar to ridge)
- **Point-laplace:** spike at zero + Laplace slab (heavier tails, similar to LASSO)

> “The Prior Learns From the Data”
>
> Instead of choosing a penalty strength λ by cross-validation, EBNM estimates the mixing proportion (how sparse?) and the scale (how large are the non-zero effects?) directly from the observed data. Implemented via the `ebnm` R package (Stephens lab, U. Chicago).

Most of you know penalised regression — LASSO, ridge — where lambda controls shrinkage. The key limitation is that lambda is fixed across all coefficients. EBNM generalises this: it estimates the prior from the data using empirical Bayes. The prior has two parts: a spike at zero (for truly zero effects) and a slab (for non-zero effects). The mixing proportion and slab scale are both estimated. Point-normal gives soft thresholding; point-laplace gives heavier tails that preserve larger effects better.

Inference: Factor-Wise CAVI

Coordinate Ascent Variational Inference

Gauss-Seidel update order (per iteration):

For each factor $k = 1, \ldots, K$:

1. **Update** $q(L_k)$ — patient loadings via EBNM
    - Precision: $A_{L,i} = \sum_j \tau_j E[F_{jk}^2] + W_{ii} E[\beta_k^2]$
    - Pseudo-data: $x_i = B_{L,i}/A_{L,i}$, scale: $s_i = 1/\sqrt{A_{L,i}}$
2. **Update** $q(F_k)$ — gene weights via EBNM (pure genomics)
3. **Update** $q(\beta_k)$ — survival coefficient via EBNM (scalar)
4. Deflate residual R_k before proceeding to $k+1$

Convergence criteria:

- $\Delta \bar{L} < 10^{-3}$ **AND** $\Delta \hat{\beta} < 10^{-3}$
- Dual criterion prevents premature stopping

Automatic K selection:

- Initialise $K_{\max} = 10$
- After convergence, prune factors with:
    - PVE $< 1\%$ (negligible expression contribution)
    - $|\hat{\beta}| \approx 0$ (no survival signal)

- Remaining factors define K_{eff}

CAVI updates one factor at a time in Gauss-Seidel order — this means each update sees the most recent values from all other factors. For each factor k, we update the loadings L, gene weights F, and survival coefficient beta, each via an EBNM call. The precision A captures how much information we have about each parameter. The auto-prune starts with 10 factors and lets the data decide how many to keep based on variance explained and survival relevance.

Seven PDAC Cohorts

Independent Validation Across Three Platforms

Dataset	Platform	n	Events	Censoring %
TCGA_PAAD	RNA-seq	144	75	47.9
CPTAC	Proteomics	129	64	50.4
Dijk	RNA-seq	90	81	10.0
Moffitt_GEO	Microarray	123	83	32.5
PACA_AU_array	Microarray	63	38	39.7
PACA_AU_seq	RNA-seq	52	31	40.4
Puleo_array	Microarray	288	181	37.2

- **RNA-seq** — mRNA transcript counts (TCGA, Dijk, PACA_AU_seq)
- **Microarray** — hybridisation arrays (Moffitt, PACA_AU_array, Puleo)
- **Proteomics** — mass spectrometry protein abundance (CPTAC)

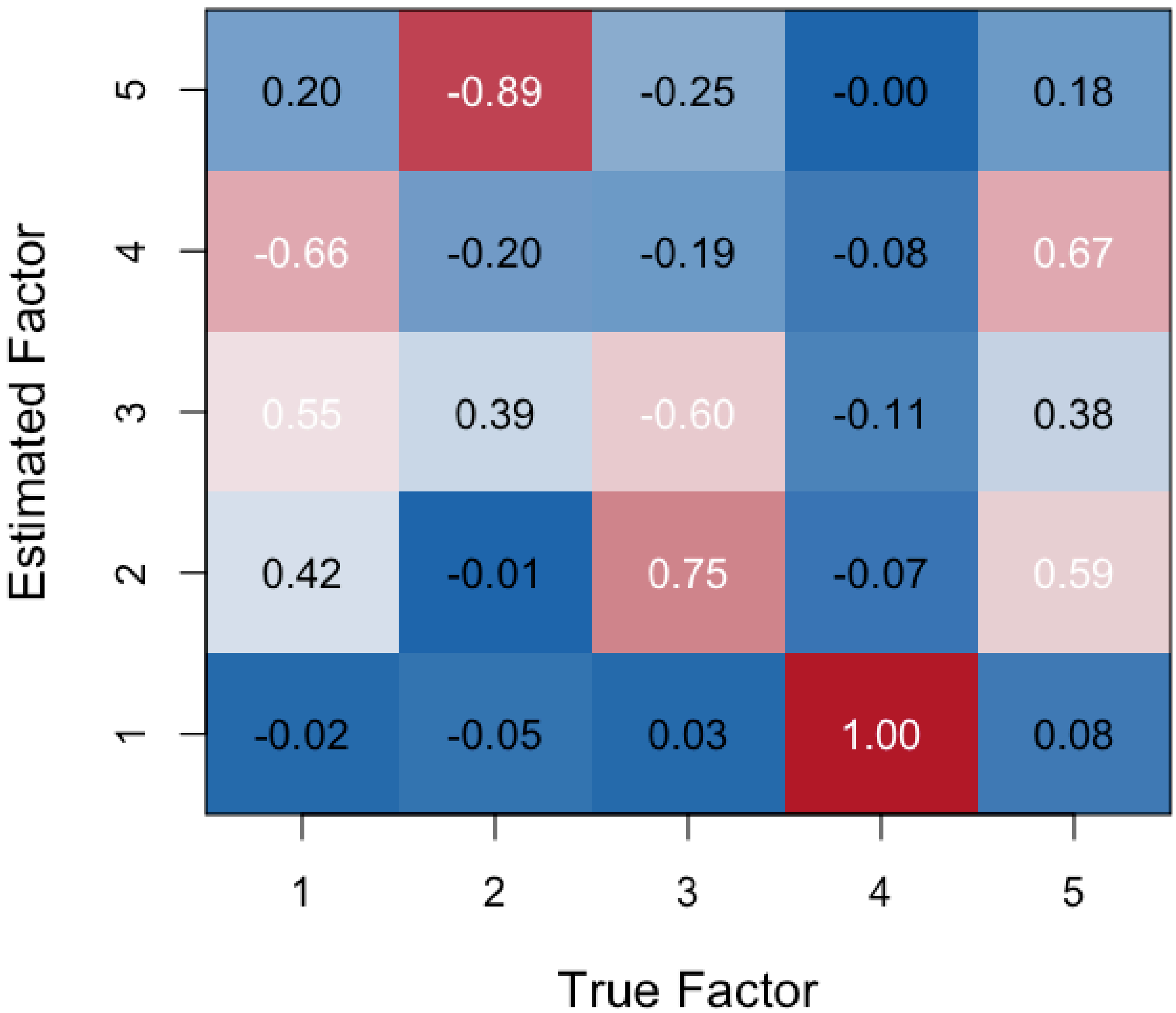
- **Why 7 datasets?** Independent validation across platforms; $n = 52$–288; top 5,000 most-variable features, column-centred

We applied the model to 7 independent PDAC cohorts spanning three platforms. This is important for showing that the method isn't overfitting to one dataset or one technology. Sample sizes range from 52 (PACA_AU_seq) to 288 (Puleo). All use the top 5000 most variable features, column-centred to remove the global mean offset before SVD initialisation.

Synthetic Validation

Known Ground Truth Recovery — $n = 200$, $p = 500$, $K_{\text{true}} = 5$

Figure 7: |Correlation| Between True and Estimated Loadings



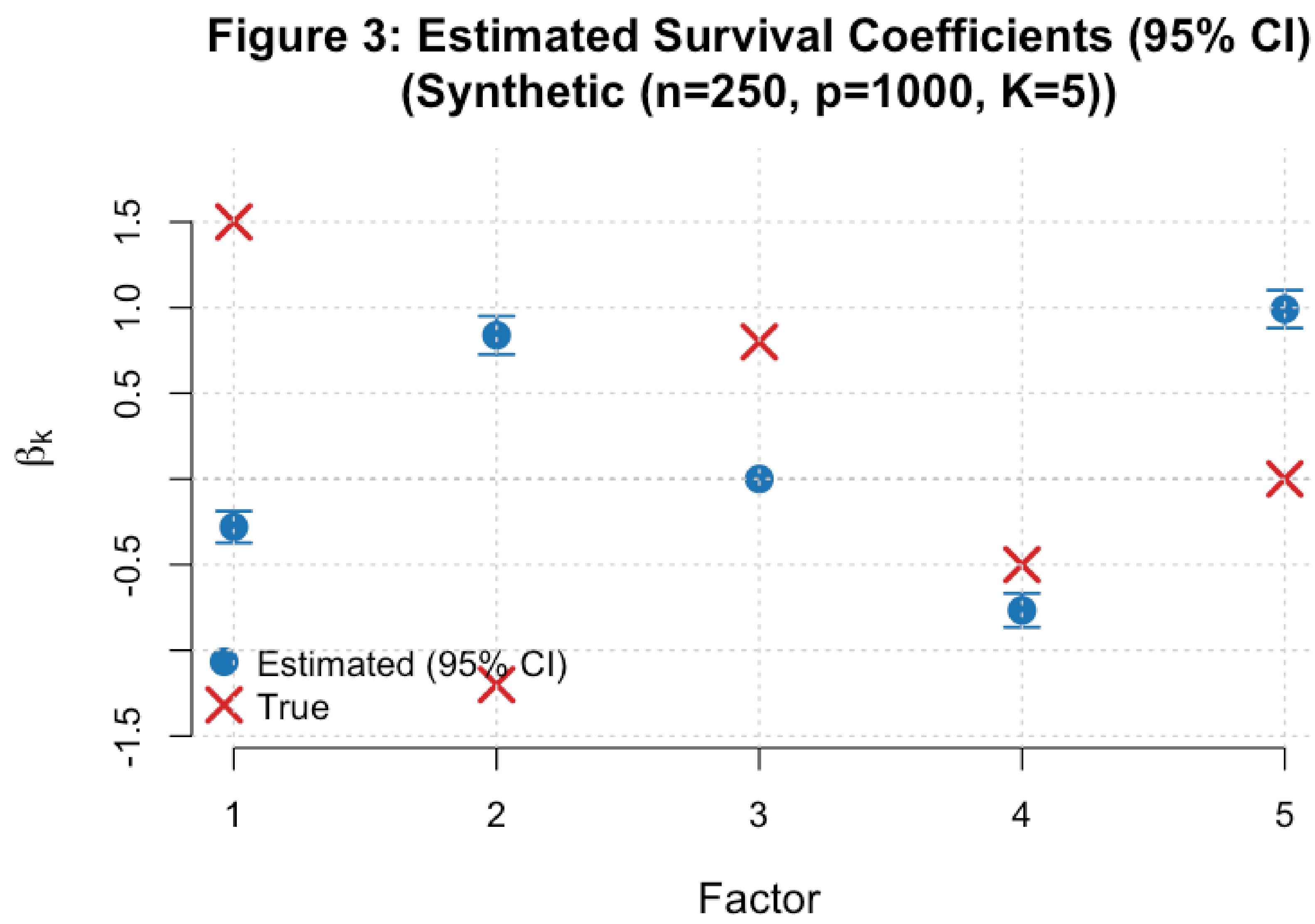


Figure 2: True vs. estimated survival coefficients $\hat{\beta}$

Factor recovery: one-to-one correspondence between true and estimated factors (up to permutation)

Null factor correctly zeroed: $\hat{\beta}_5 \to 0$ · **C-index:** Supervised 0.86 vs. PCA 0.83

Before moving to real data, we validate on synthetic data with known ground truth. The correlation matrix shows clear one-to-one correspondence between true and estimated factors. The beta comparison shows that the model recovers the correct signs and magnitudes, and critically, the null factor (true beta=0) is correctly shrunk to zero. This is the EBNM prior doing its job.

Convergence & Baseline Performance

All Seven Cohorts Converge

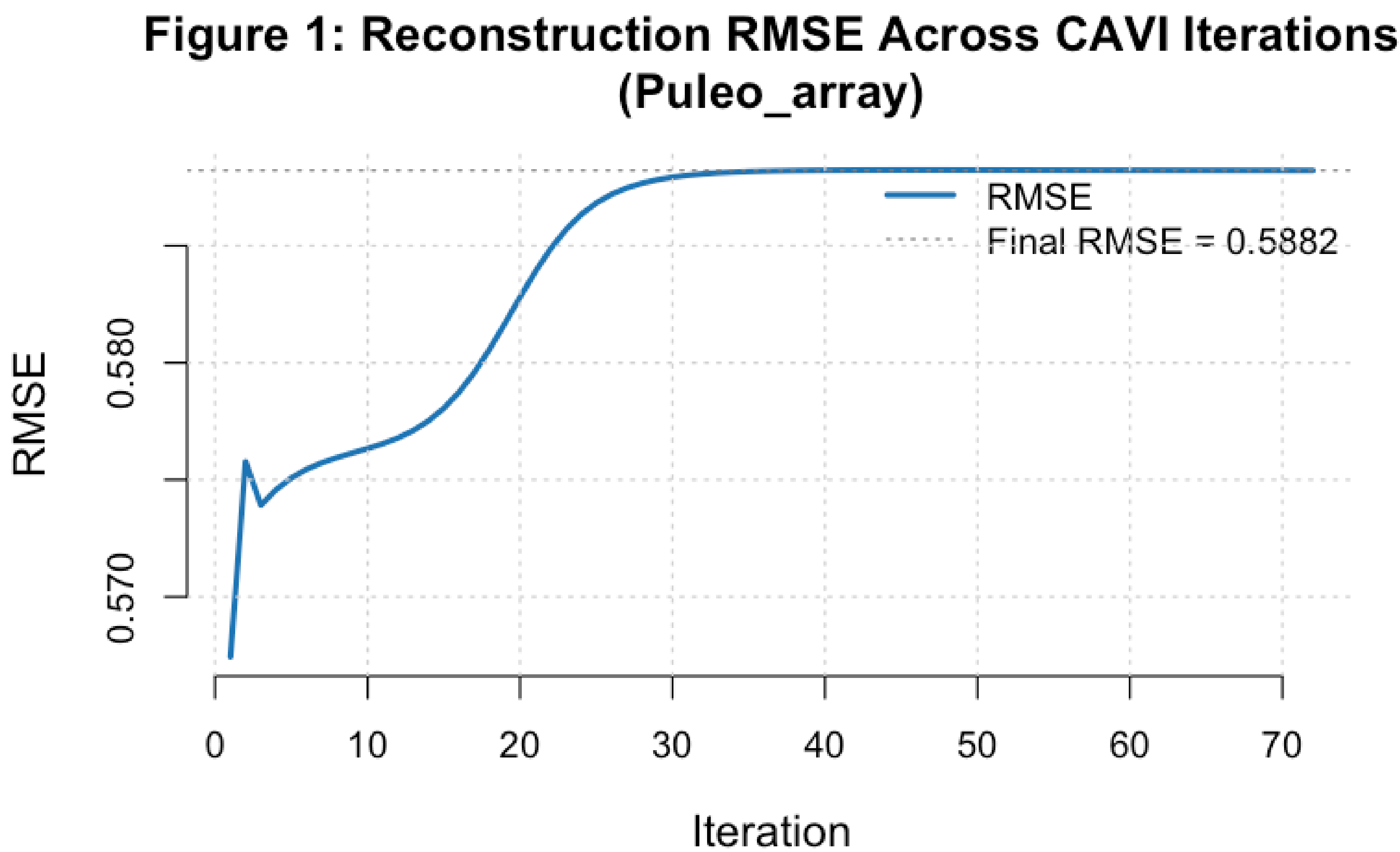


Figure 3: RMSE trace — Puleo array ($n = 288$, converges in 72 iterations)

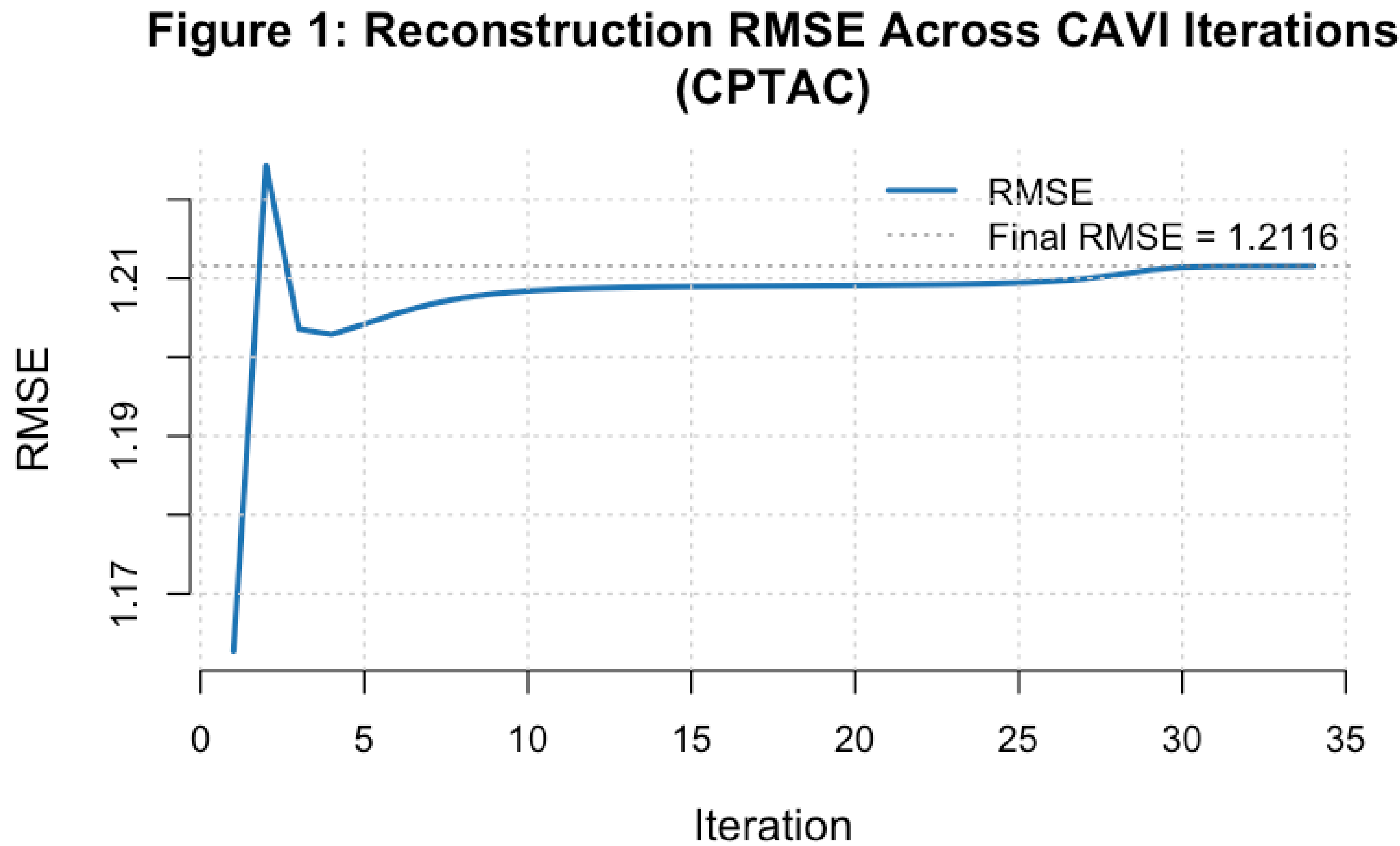


Figure 4: RMSE trace — CPTAC proteomics ($n = 129$, converges in 259 iterations)

- **All 7 cohorts converge** — RMSE descends monotonically in every case
- **Iteration range:** 14 (Moffitt, microarray) to 259 (CPTAC, proteomics)
- Microarray datasets converge fastest; proteomics requires the most iterations (higher platform noise)
- At baseline (point-normal, $K = 5$): supervised C-index $\geq$ PCA in 3/7 cohorts

RMSE descends monotonically in every cohort — the CAVI algorithm is reliably converging. Puleo is the largest cohort and converges in about 70 iterations. CPTAC proteomics needs 259 iterations, reflecting higher noise in protein abundance measurements. These baseline results use point-normal prior at K=5, which we'll see is

Dataset	K_eff	PN K=5	PL K=5	PN K_eff	PL K_eff
CPTAC	8	0.389	0.605	0.635	0.623
Dijk	10	0.381	0.483	0.544*	0.510
Moffitt_GEO_array	10	0.475	0.580*	0.530	0.550
PACA_AU_array	8	0.405	0.667	0.738	0.738
PACA_AU_seq	9	0.185	0.889	0.741	0.778
Puleo_array	10	0.396	0.594	0.674	0.649
TCGA_PAAD	9	0.341	0.707*	0.625*	0.620*

* = hit 300-iteration limit. PN = point-normal, PL = point-laplace. 80/20 stratified hold-out.

actually the worst-case configuration.

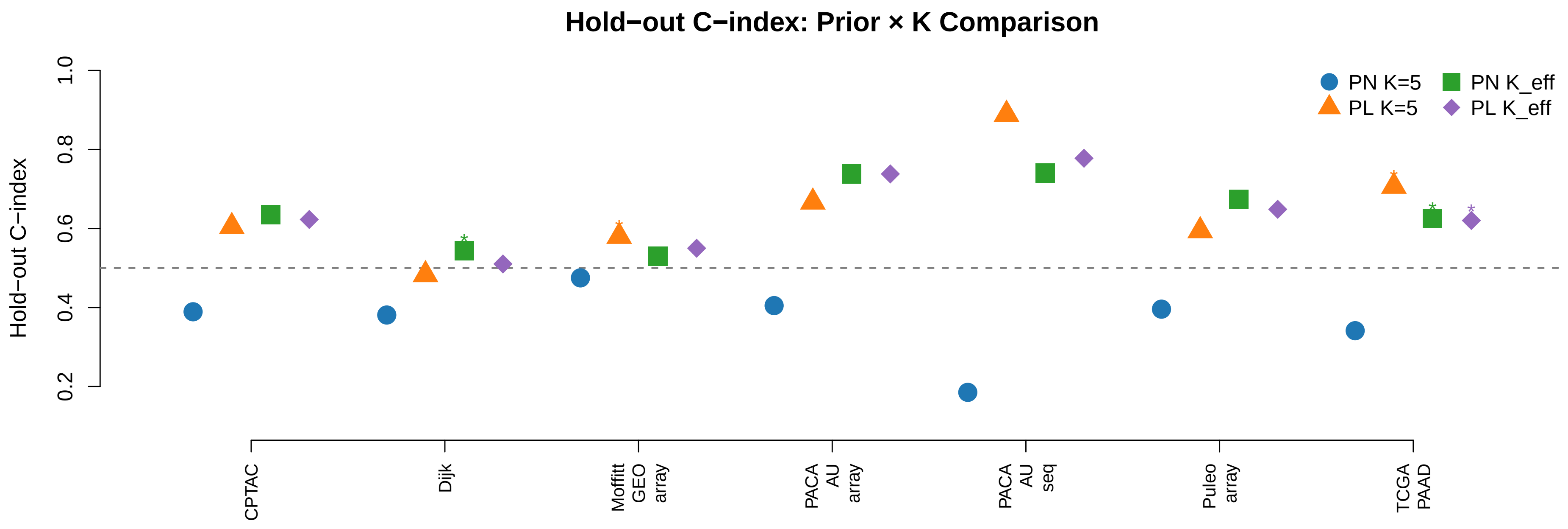
Prior × K Factorial: The Central Result

Hold-Out C-index Across Four Conditions

- **Hold-out C-index** — concordance between predicted risk scores and observed survival on 20% held-out patients (random chance = 0.50; higher is better)
- **PN K=5** frequently falls below 0.50 (degenerate solutions); **PL K=5** reliably improves over point-normal
- **K_eff** conditions (both PN and PL) consistently outperform fixed $K = 5$ — K selection is the dominant lever

This is the central results table. Four conditions per dataset: point-normal or point-laplace prior, at fixed K=5 or data-driven K_eff. Two patterns jump out: point-normal at K=5 often gives C-index below chance (0.19 for PACA_AU_seq, 0.34 for TCGA) — these are degenerate solutions where beta collapses. And the K_eff conditions consistently outperform K=5 regardless of prior.

Prior × K: Scatter Plot



- Each point = one dataset × one condition; dashed line = random chance (C = 0.50); four colours = the four prior × K conditions
- **Blue circles (PN K=5):** several datasets fall well below chance — degenerate fits where $\hat{\beta} \to 0$
- **Moving right (PL → K_eff):** progressive improvement; K_eff conditions (green/purple) cluster above the dashed line across all cohorts

The scatter plot makes the pattern visual. Blue circles at the bottom are the degenerate point-normal K=5 fits. Orange triangles show the improvement from switching to point-laplace. But the biggest jump is from K=5 to K_eff — the green and purple points are consistently higher regardless of prior.

Key Findings: Prior × K Comparison

Point-laplace preferred at fixed K

- Point-laplace **outperforms** point-normal in **all 7 cohorts** at $K = 5$
- ΔC_{test} from +0.06 (Moffitt) to +0.70 (PACA_AU_seq)
- Point-normal at $K = 5$ degenerates in 4/7 cohorts ($C < 0.4$)
- Heavier tails better capture continuous, non-zero gene weight structure

K selection dominates prior choice

- $K_{\text{eff}} \in \{8, 9, 10\}$ — **always above** $K = 5$; every K_{eff} condition outperforms its $K = 5$ counterpart
- At K_{eff}: prior advantage is **mixed** (PN wins 4/7)
- Three cohorts saturate at $K_{\max} = 10$

  Takeaway: invest in K selection, not prior tuning.

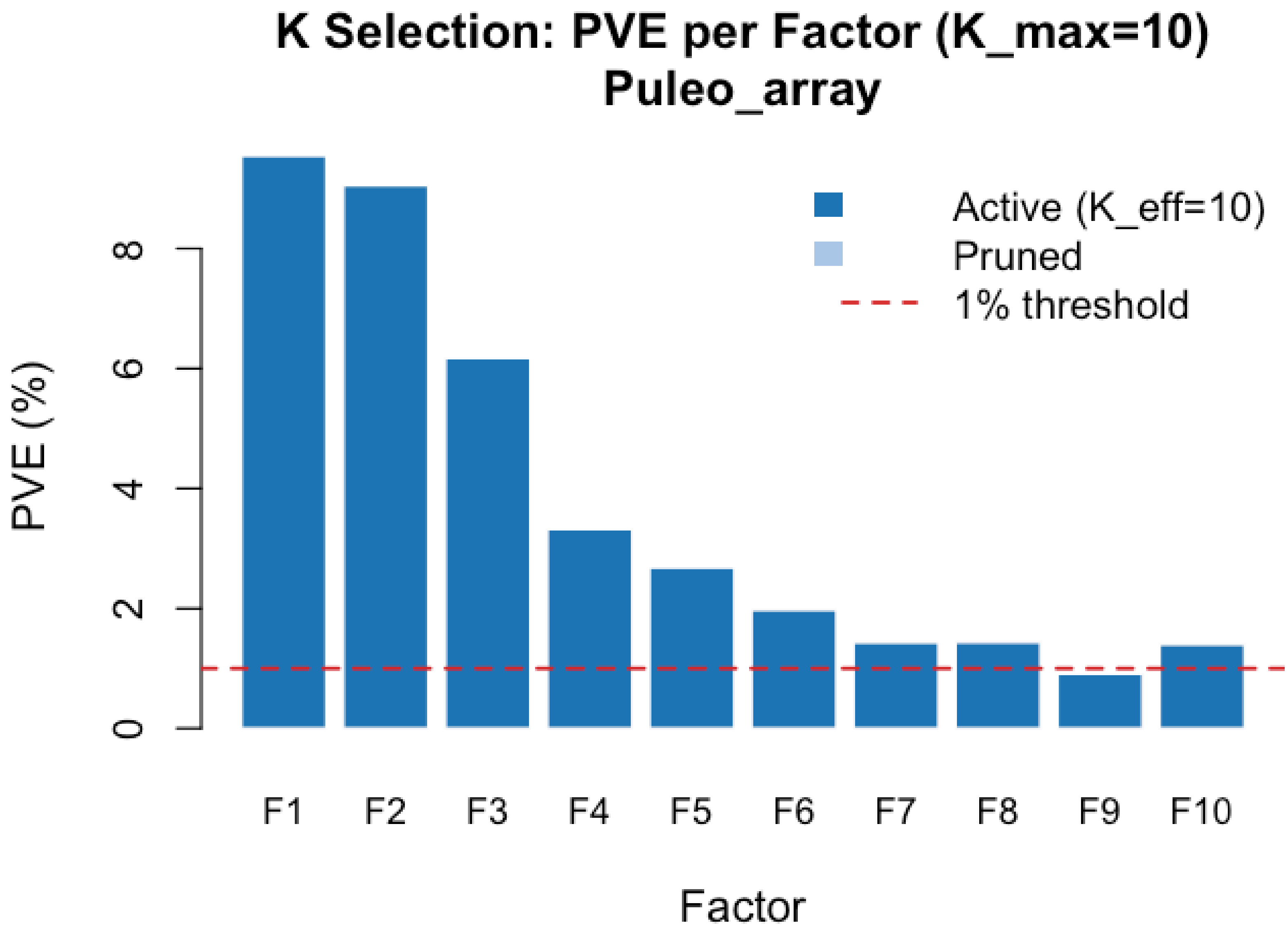


Figure 5: PVE scree — Puleo ($K_{\text{eff}} = 10$)

Two main findings. First, point-laplace is clearly better than point-normal at K=5. The Gaussian prior over-shrinks in real data. But the second finding is more important: K selection dominates everything. Once you give the model enough factors, the prior advantage largely disappears. The scree plot shows Puleo needing 10 factors — our initial guess of 5 was way too conservative. Three cohorts actually saturate at our K_max of 10, meaning there may be even more structure to discover.

Deep Dive: Puleo Array ($n = 288$)

The Clearest Factor Structure

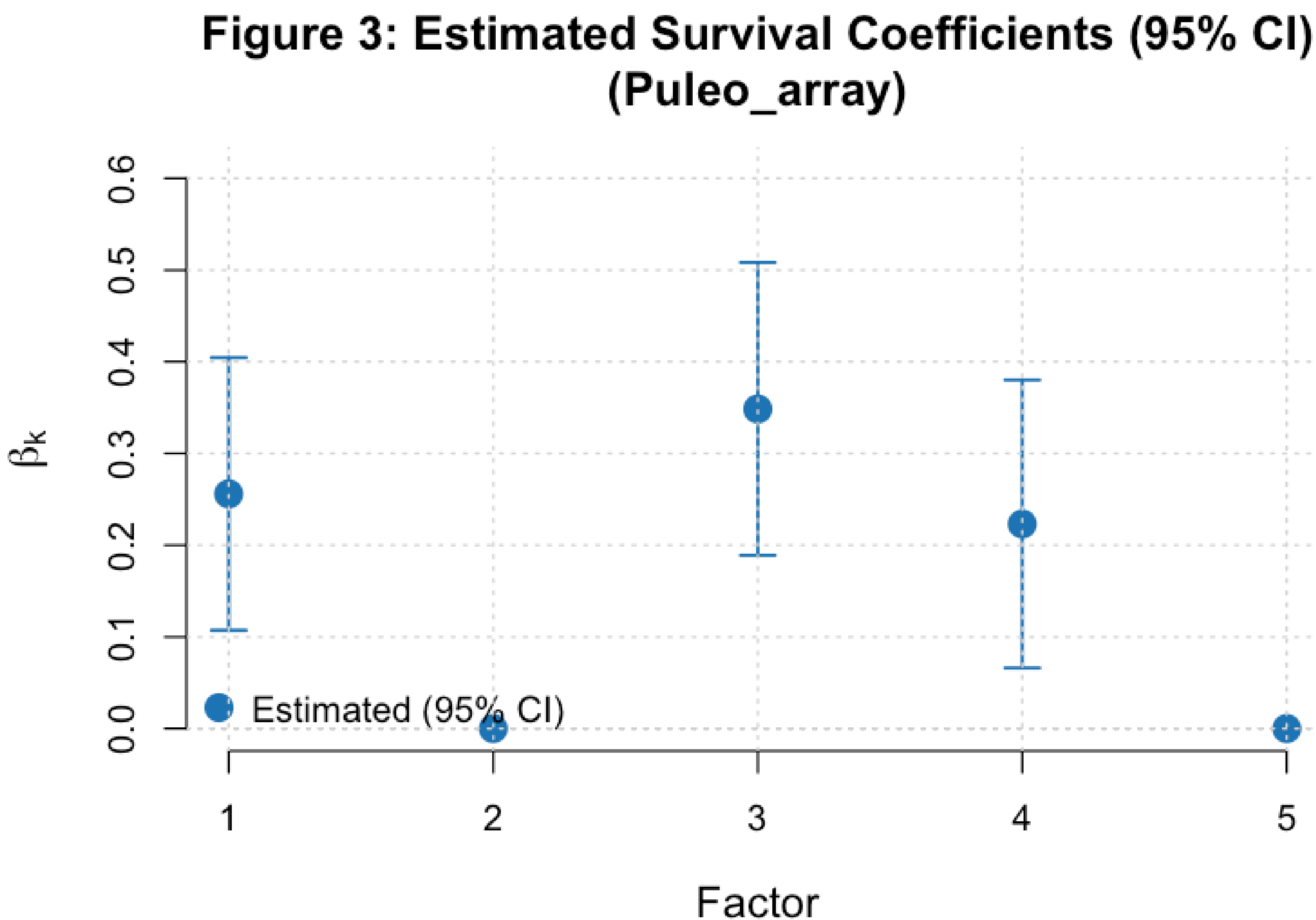


Figure 6: Survival coefficients $\hat{\beta}$ per GEP

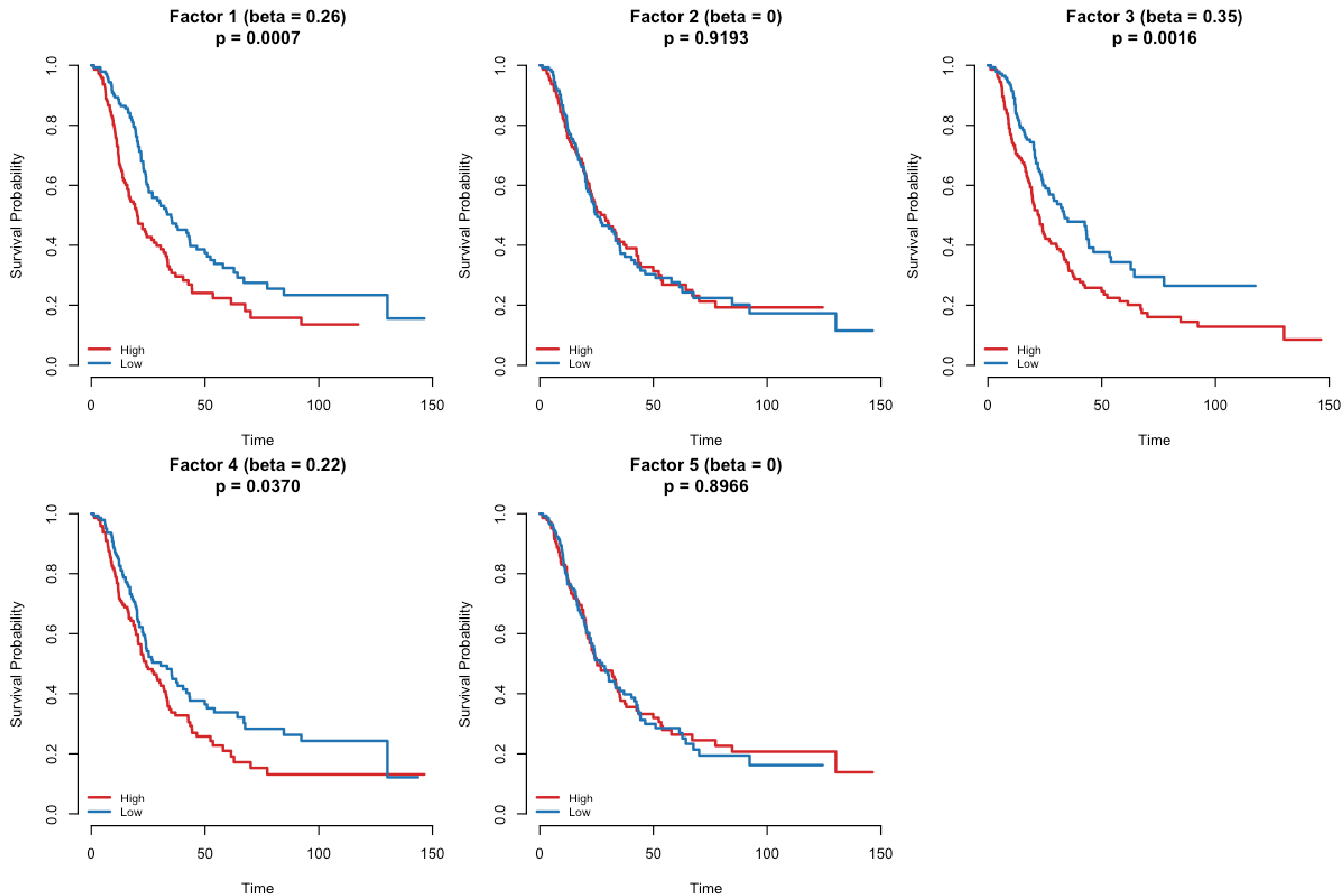


Figure 7: Kaplan-Meier survival curves by GEP activation

- **Factor 2:** highest PVE (9.05%) but $\hat{\beta} = 0$ — expression-dominant, survival-neutral
- **Factors 3, 4, 9:** large $|\hat{\beta}|$ — prognostically informative
- The model **separates prognostic from non-prognostic** structure — PCA would test all factors indiscriminately
- $K_{\text{eff}} = 10$ via auto-prune (4 more factors than fixed $K = 5$)

Puleo is our best exemplar — largest cohort, clearest structure. The beta barplot shows factor 2 is shrunk to zero despite being the single largest expression axis. A post-hoc approach would have tested it alongside the genuinely prognostic factors, diluting statistical power. The KM curves show clear patient stratification by GEP activation. This is what supervised factorisation buys you.

Deep Dive: TCGA_PAAD

The Most Widely Studied PDAC Cohort

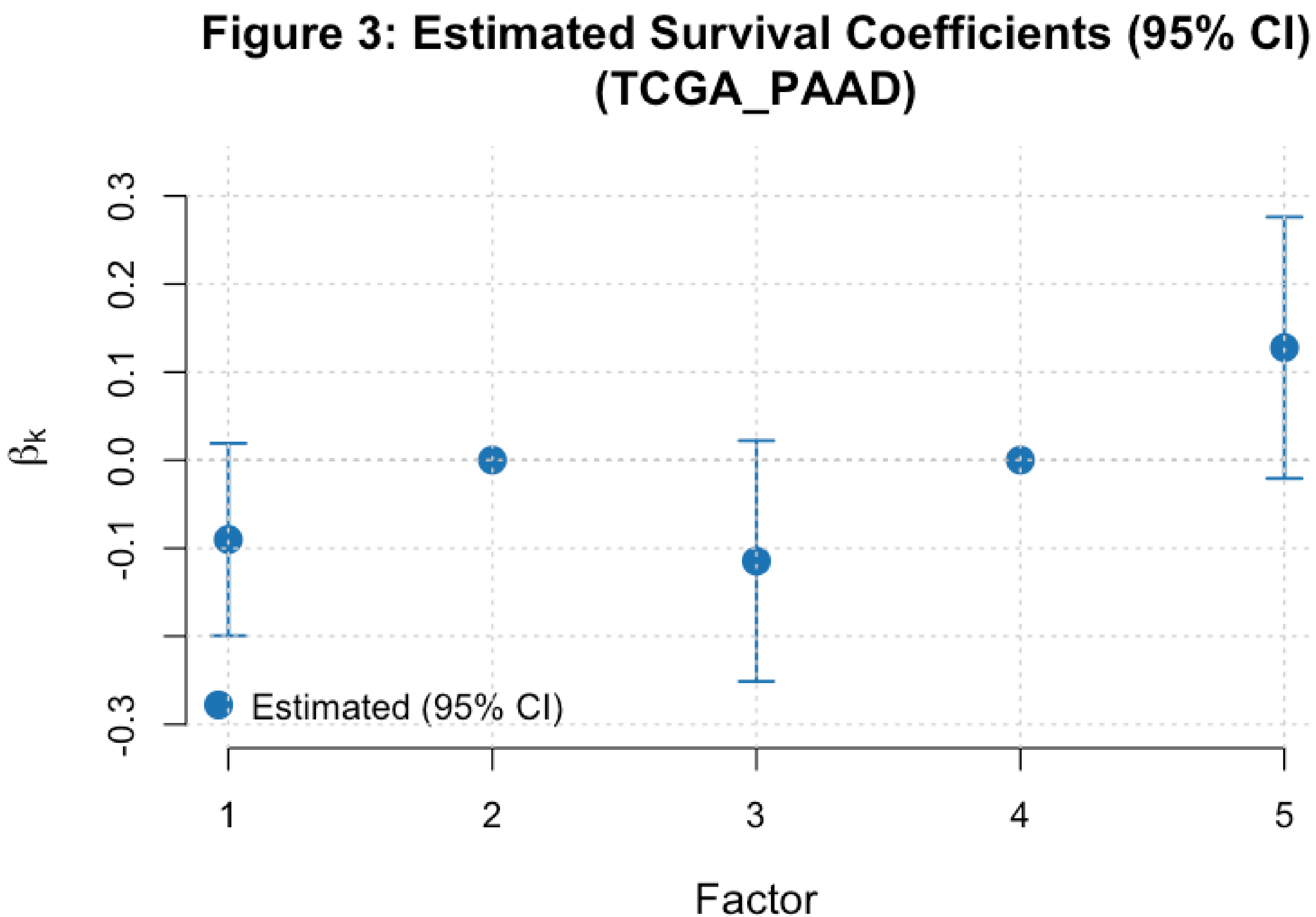


Figure 8: Survival coefficients $\hat{\beta}$ per GEP

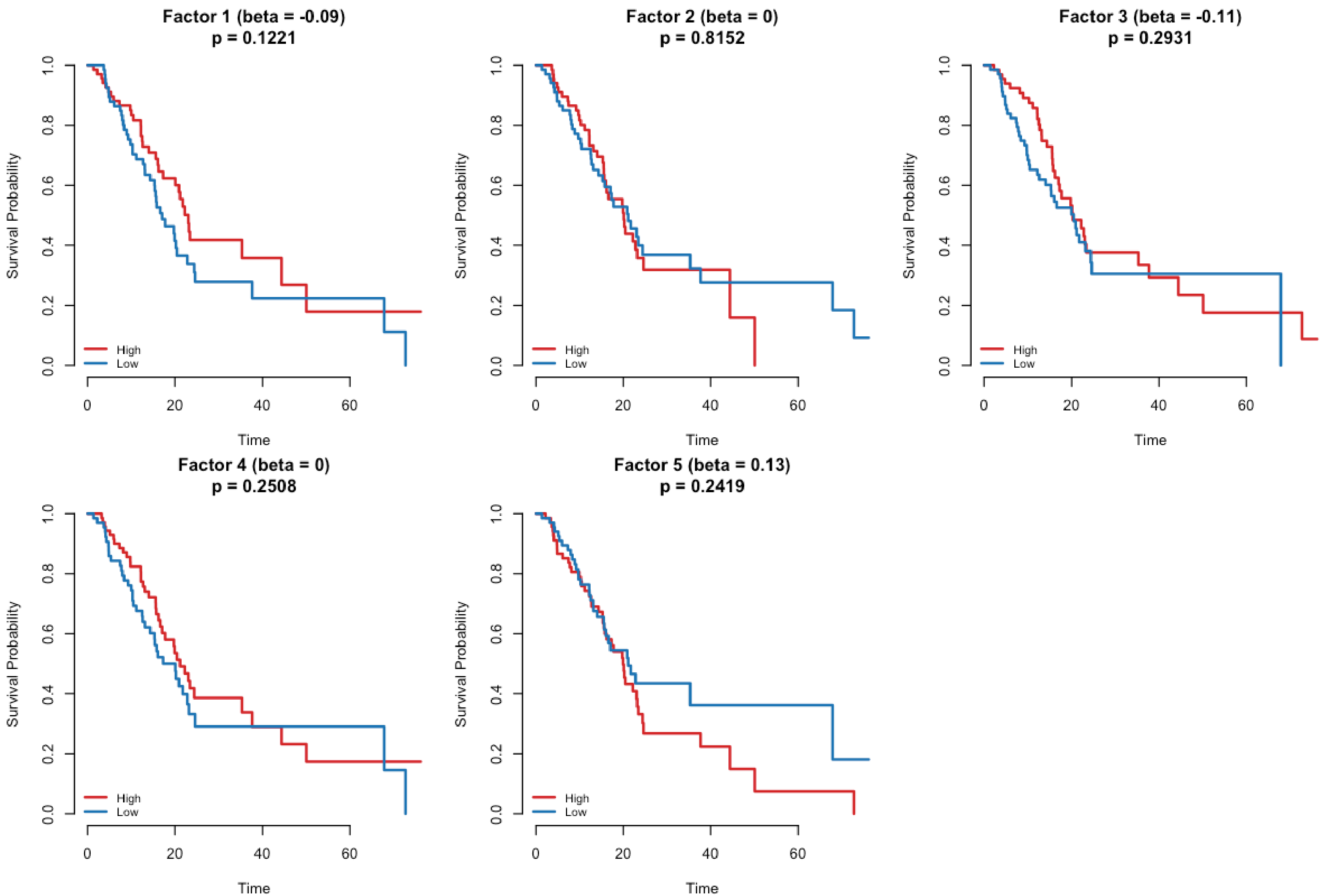


Figure 9: Kaplan-Meier survival curves by GEP activation

- **TCGA_PAAD** ($n = 144$, RNA-seq) — the benchmark PDAC dataset
- $K_{\text{eff}} = 9$; largest K_{eff} improvement: $\Delta C_{\text{test}} = +0.28$ over $K = 5$ baseline
- Consistent with Puleo: model identifies both prognostic and expression-only GEPs
- Patient stratification visible in Kaplan-Meier curves

Dataset	C (PCA)	Best Condition	K	C (SBMF)	ΔC
CPTAC	0.703	point_normal (K_eff)	8	0.635	**-0.068**
Dijk	0.610	point_normal (K_eff)	10	0.544	**-0.066**
Moffitt_GEO_array	0.571	point_laplace	5	0.580	**0.009**
PACA_AU_array	0.653	point_normal (K_eff)	8	0.738	**0.085**
PACA_AU_seq	0.776	point_laplace	5	0.889	**0.113**
Puleo_array	0.664	point_normal (K_eff)	10	0.674	**0.010**
TCGA_PAAD	0.640	point_laplace	5	0.707	**0.067**

TCGA is the dataset most people in the room will have heard of. It shows the same pattern as Puleo: some factors are prognostic, others are expression-only. The K_eff improvement here is dramatic — going from K=5 to K=9 improves hold-out C-index by 0.28. This cohort is also where we see the most dramatic failure of point-normal at K=5 (C-index of 0.34 — below random).

Hold-Out Prediction

Generalisation to Unseen Patients

- SBMF **beats PCA baseline on held-out patients in 5/7** cohorts
- Improvements most reliable when test set is large enough ($n_{\text{test}} \geq 20$)
- **Caveat:** small test sets ($n \leq 15$) produce noisy C-index estimates

On held-out data, SBMF generalises better than PCA in 5 of 7 cohorts. The improvements are most reliable in larger test sets. PACA_AU_seq has only about 10 test patients — that's too few for a reliable C-index estimate. The key message is that the model is learning genuine prognostic signal, not just overfitting to the training data.

Future Directions & Summary

Next Steps

1. **Gene set enrichment** — fgsea / Enrichr on top-loaded genes per GEP (MSigDB, KEGG, Reactome)
2. **CV-based K selection** — cross-validated C-index on Longleaf HPC ($K_{\max} \geq 15$)
3. **Multi-dataset shared factors** — identify GEPs replicating across cohorts

4. **Full ELBO as model selection** — compare models by variational lower bound
5. **Manuscript preparation**

Summary

1. **SBMF** jointly learns gene expression programs and survival associations via Bayesian inference
2. **K selection is the dominant lever** — $K_{\text{eff}} \in \{8, 9, 10\}$ vs. fixed $K = 5$
3. **Point-laplace preferred** when K is constrained
4. **Generalises** to held-out patients across 3 platforms (5/7 cohorts)
5. **Codebase:** 139/139 tests passing, modular architecture

The most pressing next step is gene set enrichment to give biological names to the statistical factors. Second, proper cross-validated K selection on our HPC cluster with K_max of 15 or higher. Third, finding which GEPs replicate across independent cohorts would be the strongest biological validation. To summarise: the method works, converges reliably across platforms, and the main insight is that getting K right matters more than prior selection.

Discussion

Questions, suggestions, and feedback welcome!

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Open the floor for discussion. Happy to go deeper on any part of the methods, results, or future plans. I have backup slides with additional per-cohort results, ELBO traces, and technical details on EBNM.

Pooled RNA-seq Analysis

Cross-Cohort Validation: TCGA + Dijk + PACA_AU_seq

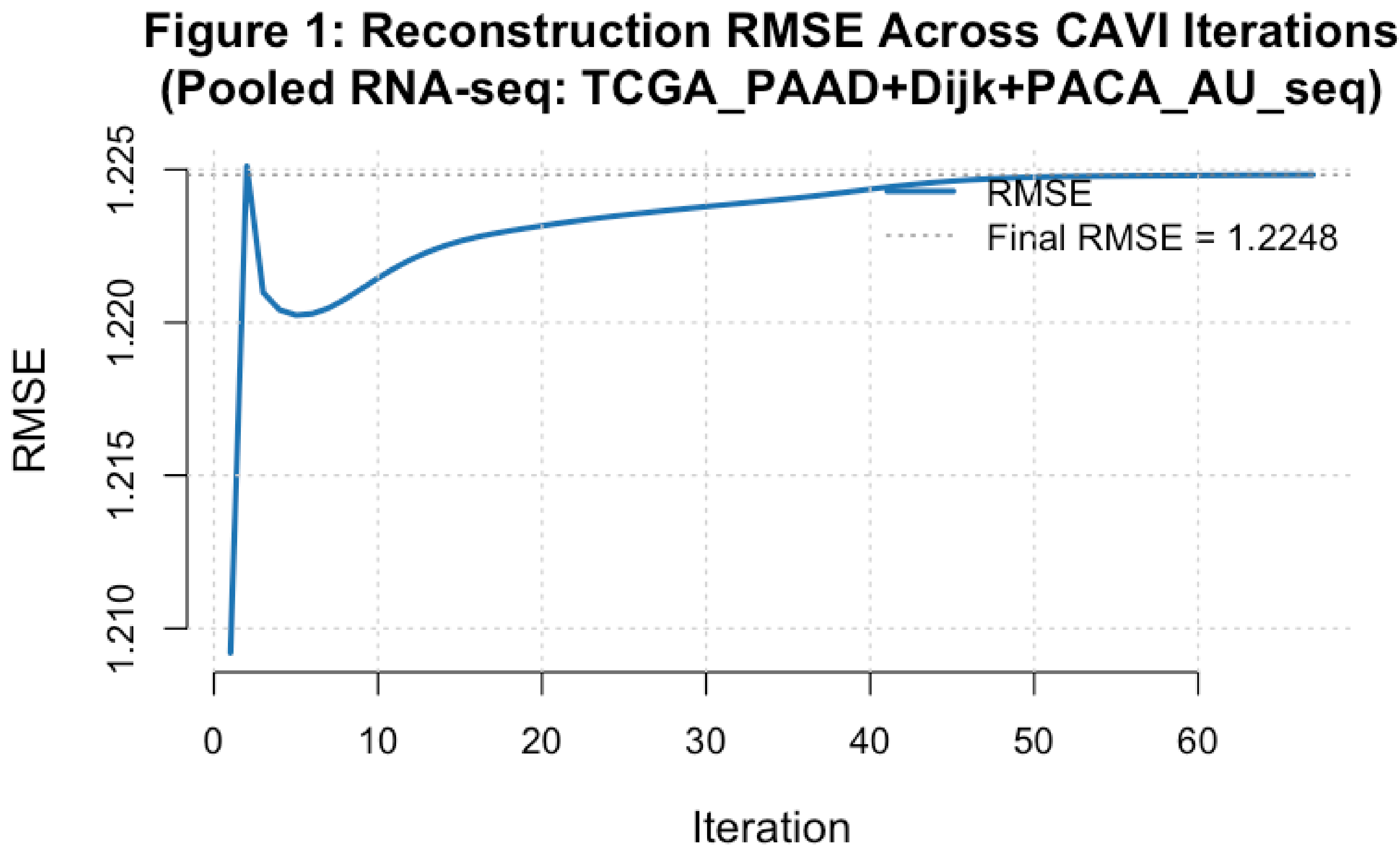

Figure 10: RMSE convergence — pooled RNA-seq ($n = 286$)

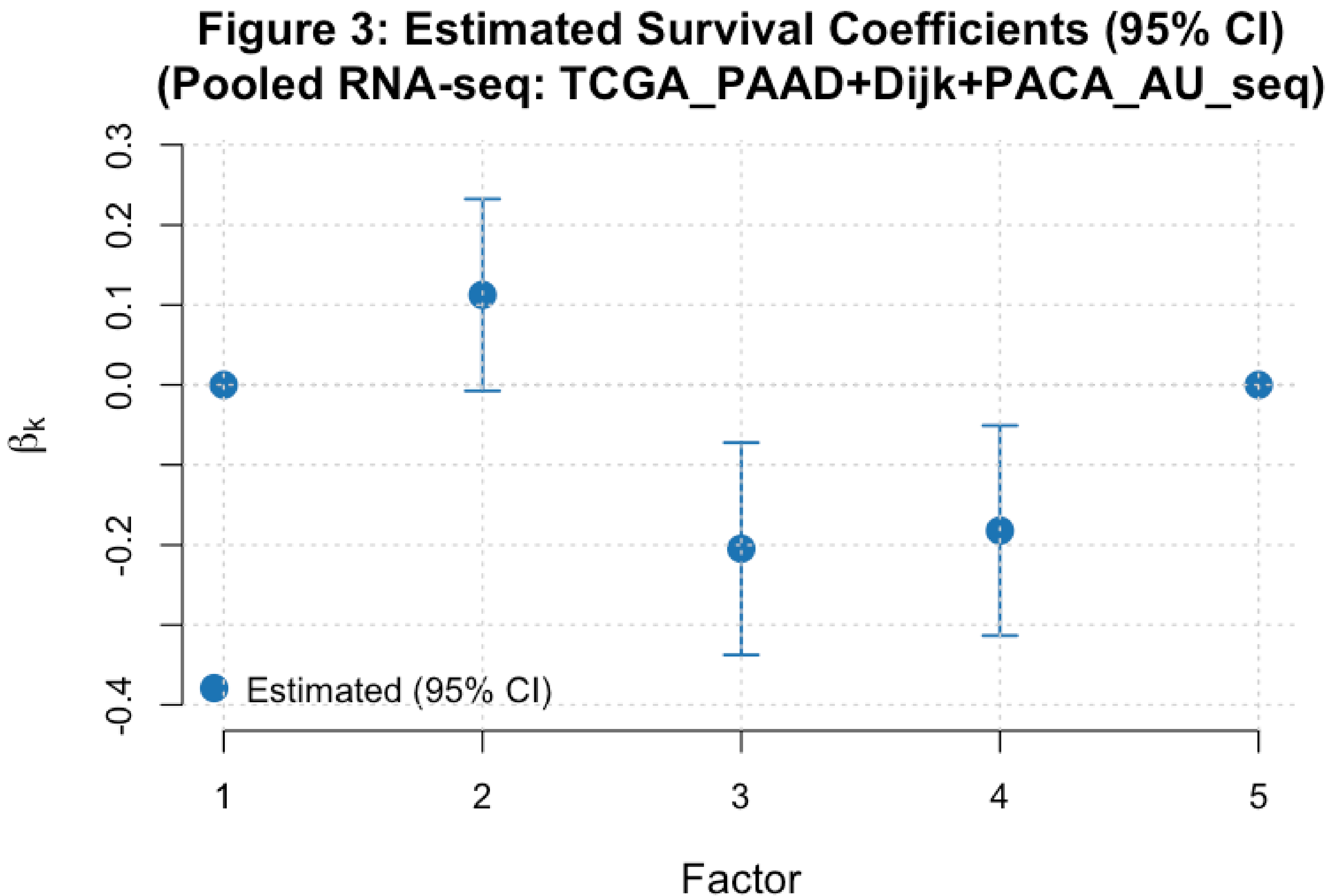

Figure 11: Survival coefficients $\hat{\beta}$ — pooled

- **Pooled:** $n = 286$ patients, 1,573 common genes (intersection across 3 RNA-seq cohorts)
- Converges in 67 iterations
- Tests whether gene programs replicate when patients from different studies are combined

EBNM Deep Dive

Empirical Bayes Normal Means — Full Detail

Each CAVI update reduces to an EBNM sub-problem:

$$x_j = \theta_j + \epsilon_j, \quad \epsilon_j \sim N(0, s_j^2)$$

where $x_j = B_j/A_j$ (pseudo-data) and $s_j = 1/\sqrt{A_j}$ (scale).

EBNM solves:

$$\hat{g} = \arg\max_{g \in \mathcal{G}} \sum_j \log \int N(x_j; \theta, s_j^2)\, g(\theta)\, d\theta$$

Then computes posterior: $E[\theta_j \mid x_j, \hat{g}]$ and $\mathrm{Var}(\theta_j \mid x_j, \hat{g})$.

Critical: Posterior second moments $E[\theta^2] = \mathrm{Var}(\theta) + E[\theta]^2 \neq E[\theta]^2$

This prevents overfitting to uncertain parameter estimates — the error-in-variables correction.

GEP Heatmap: Puleo Array

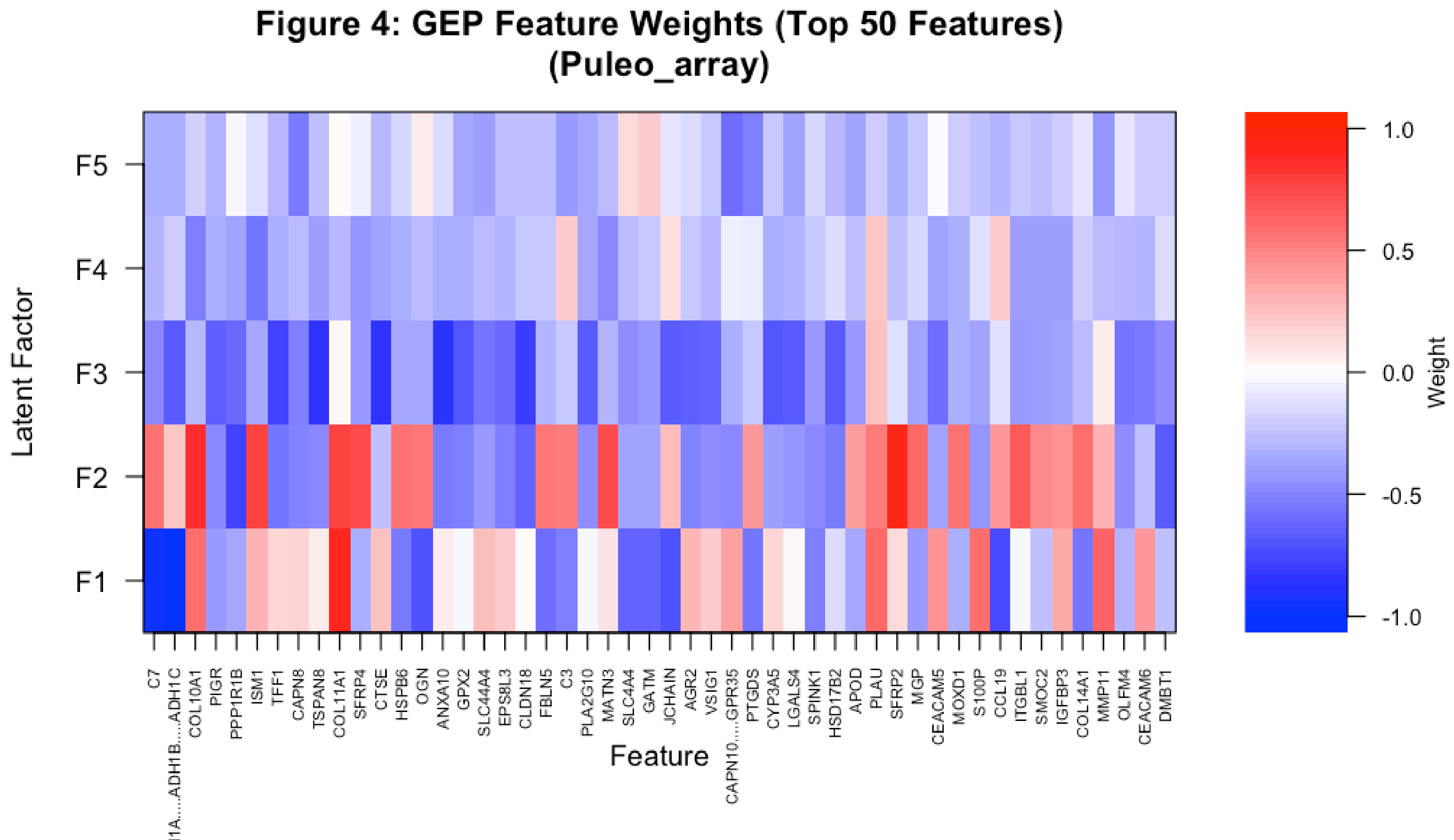

Figure 12: Gene expression program heatmap — patient loadings (rows) × GEPs (columns)

Each column is a gene expression program. Distinct patient groups emerge from the loading patterns.

ELBO Convergence Traces

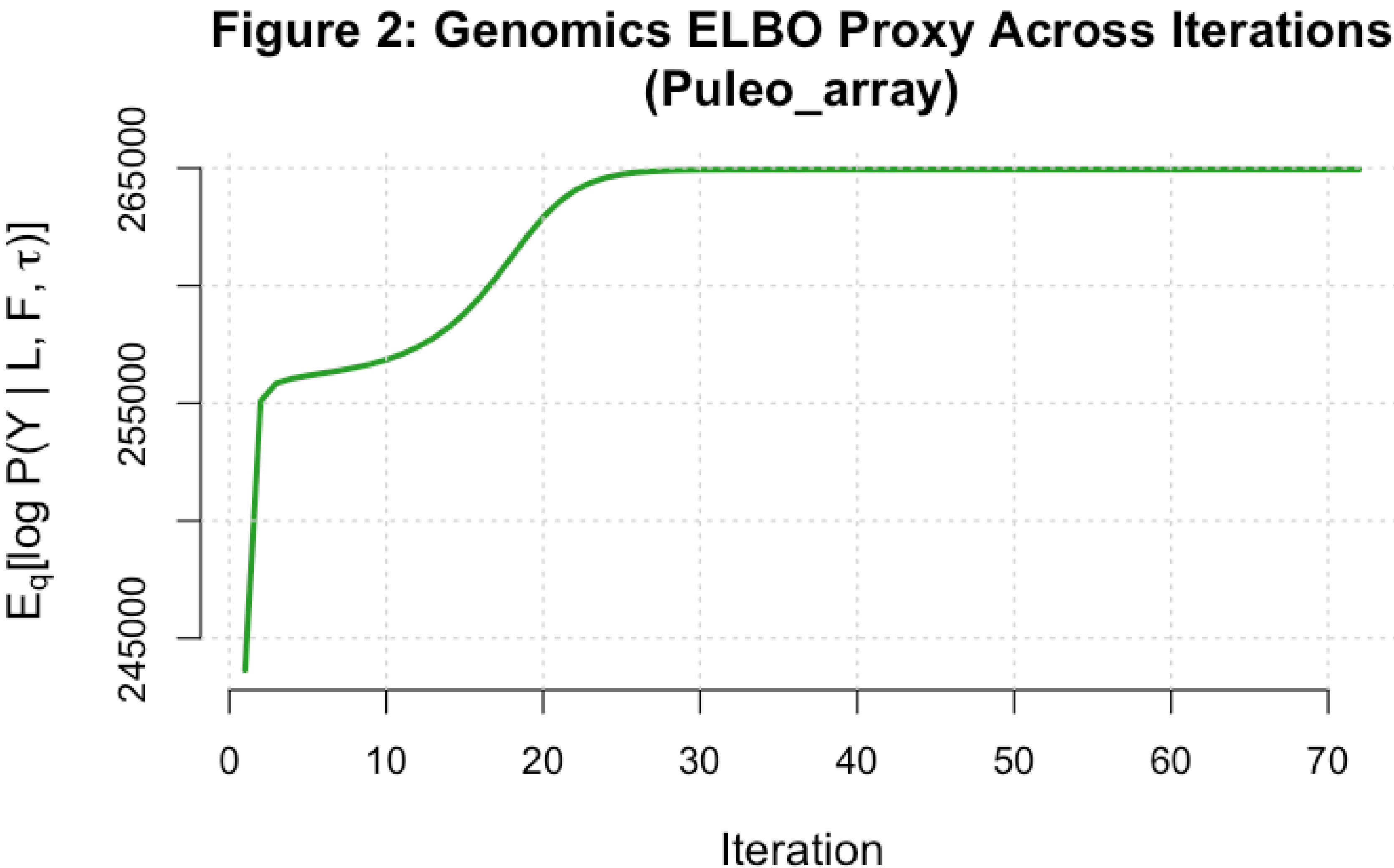

Figure 13: ELBO trace — Puleo array

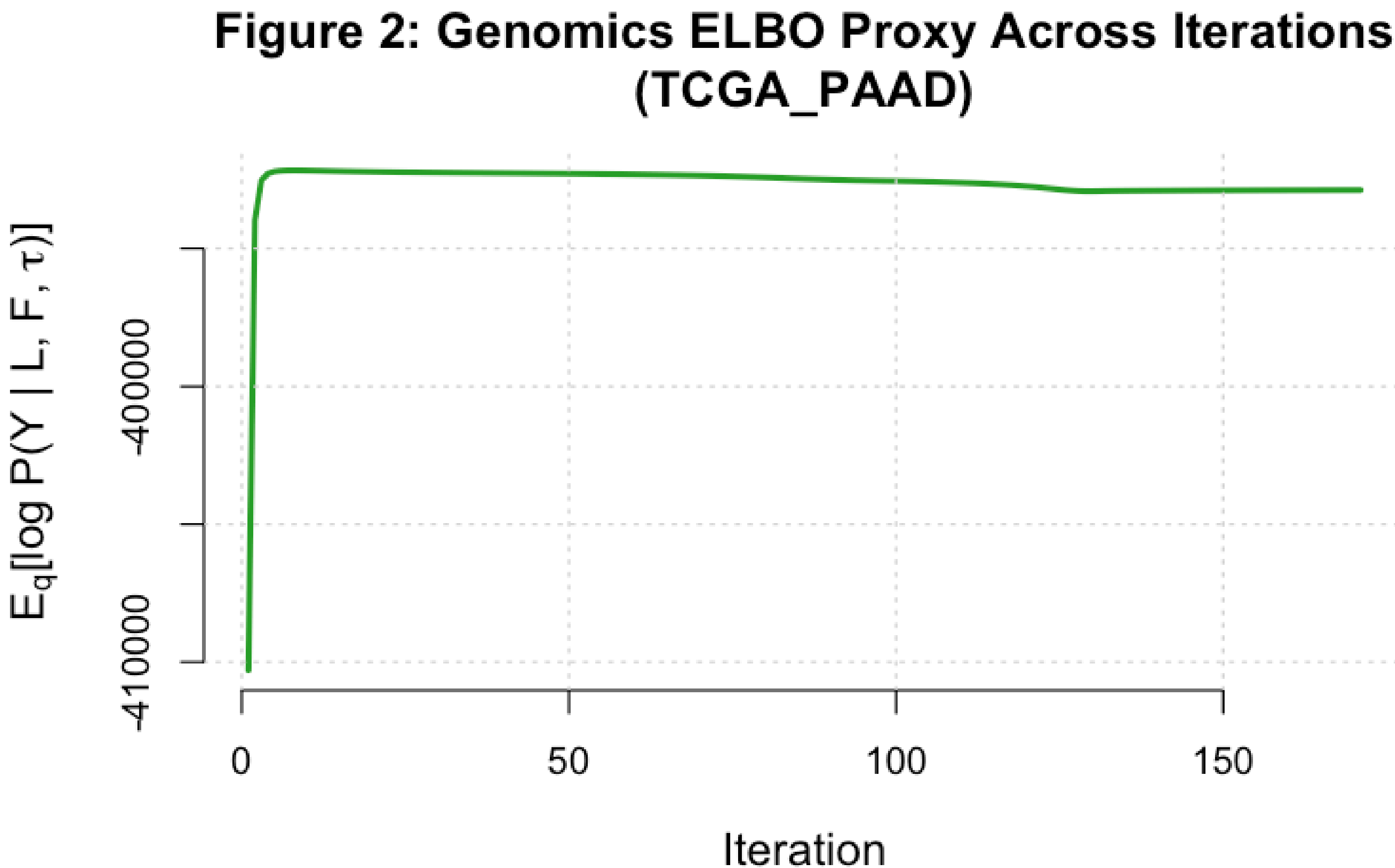

Figure 14: ELBO trace — TCGA_PAAD

ELBO (variational lower bound) increases monotonically, confirming valid CAVI convergence.

Tau/Precision Distribution

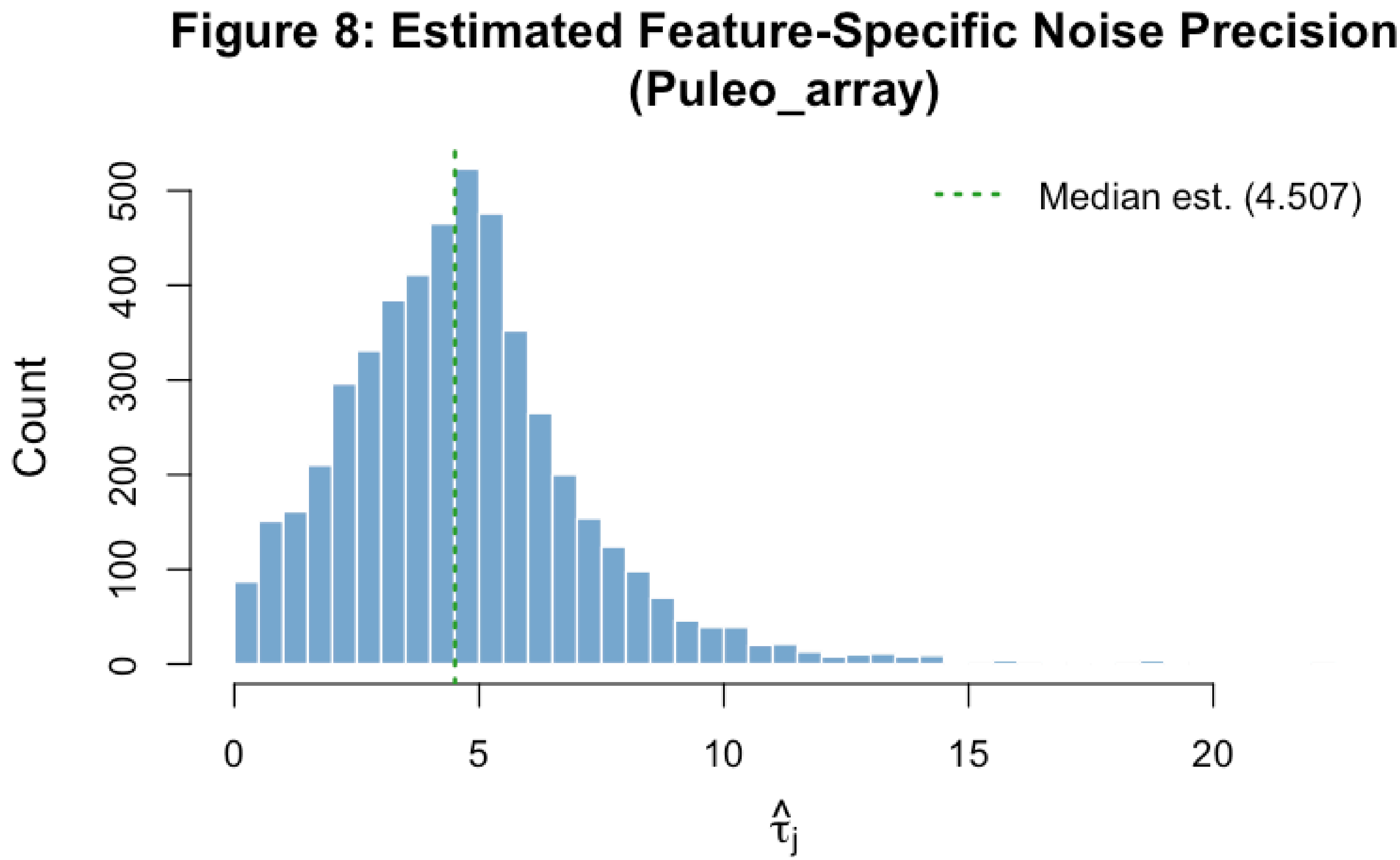

Figure 15: Gene-specific precision $\hat{\tau}_j$ — Puleo array

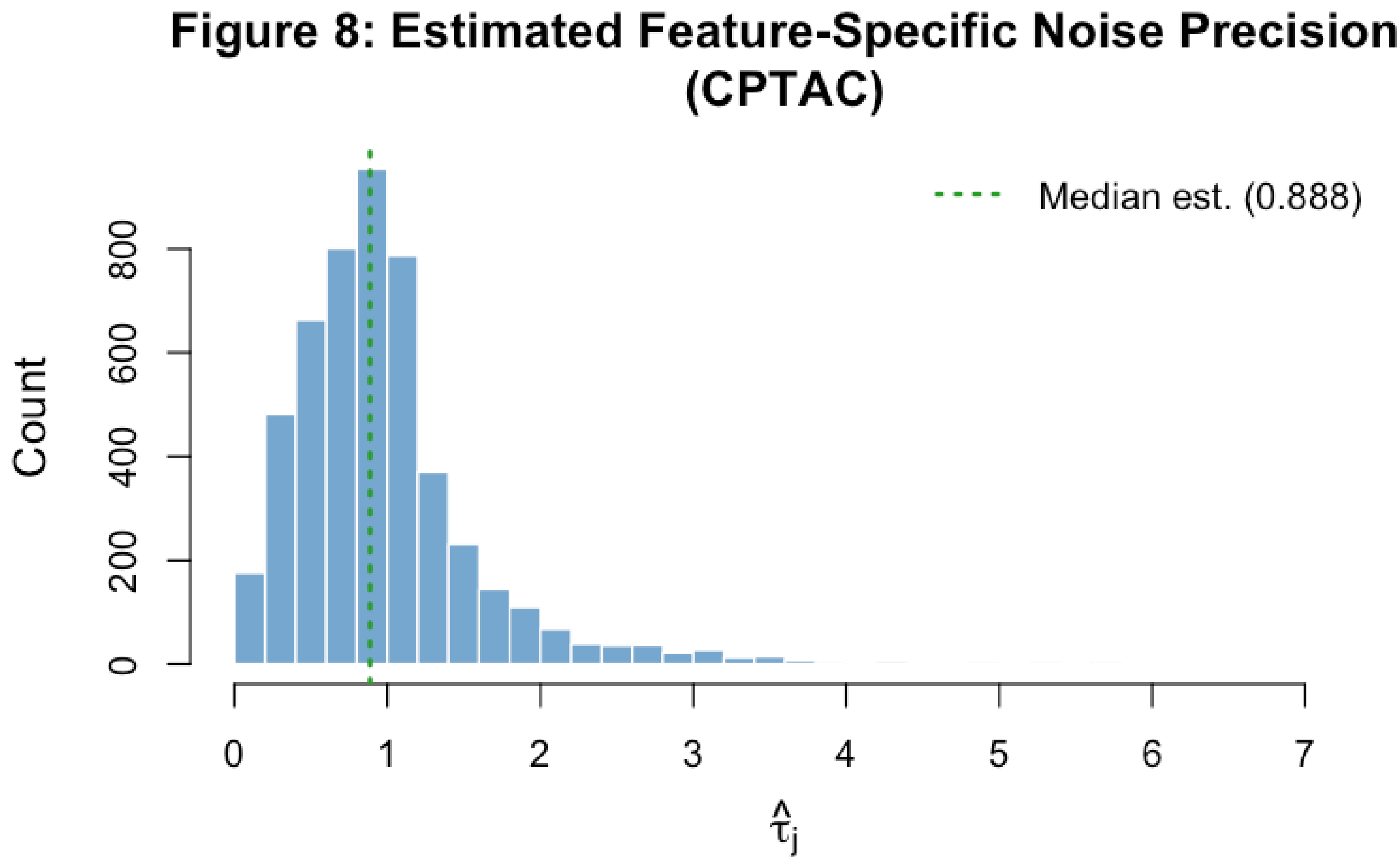

Figure 16: Gene-specific precision $\hat{\tau}_j$ — CPTAC

Heteroscedastic noise model: each gene has its own precision estimate. Some genes are precisely measured (high τ), others are noisy (low τ).

Per-Cohort Beta Comparisons

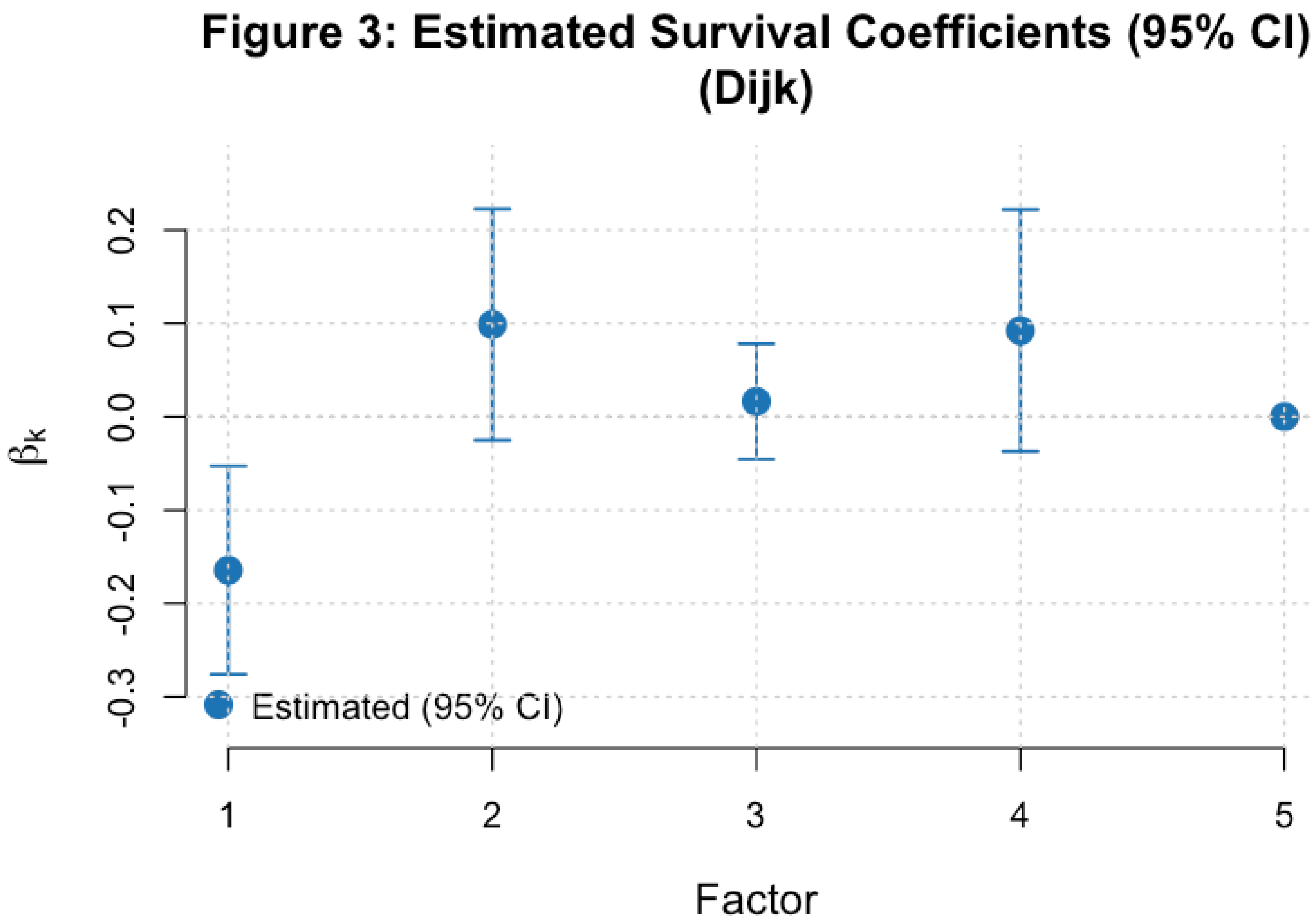

Figure 17: Dijk

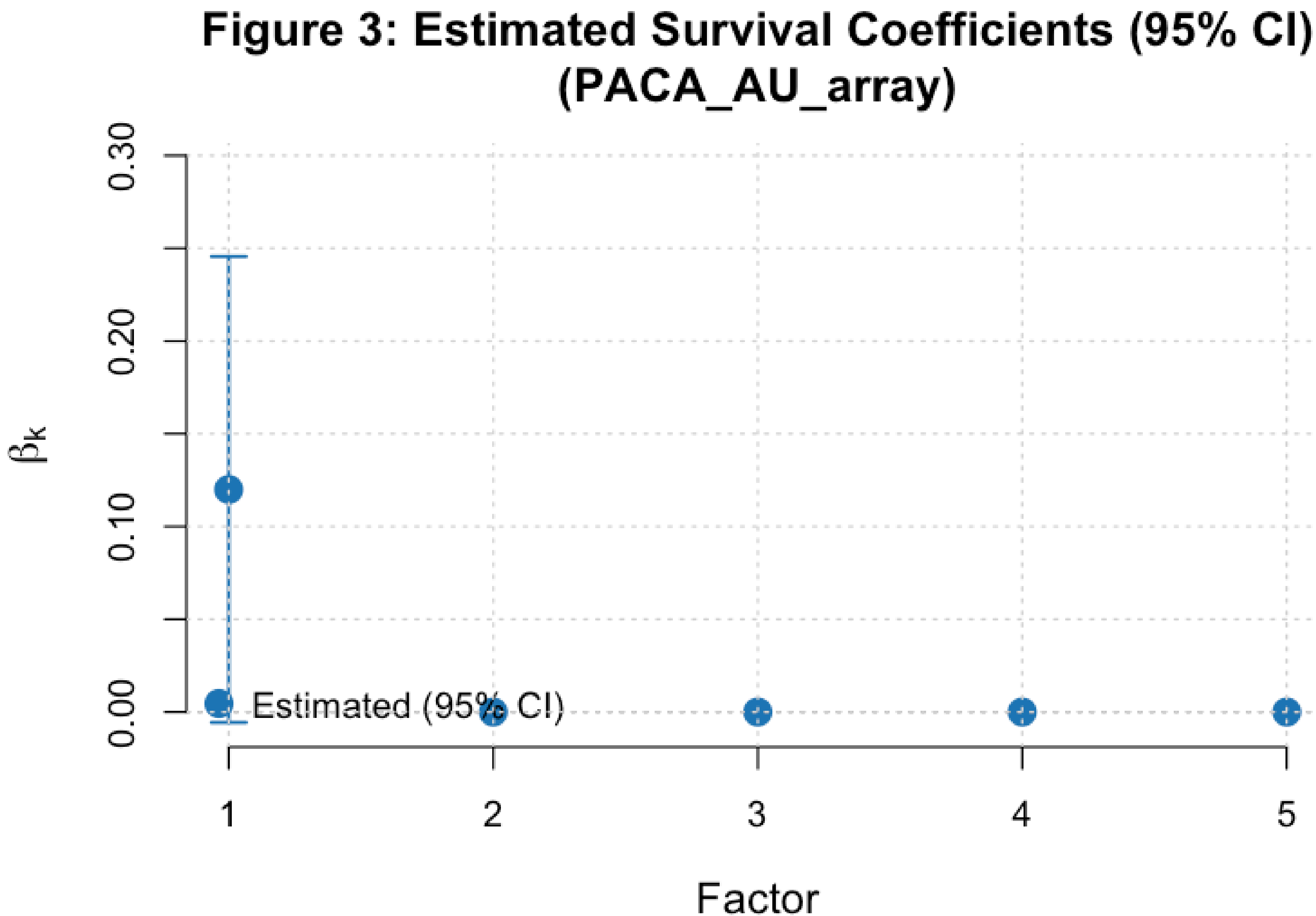

Figure 18: PACA__AU__array

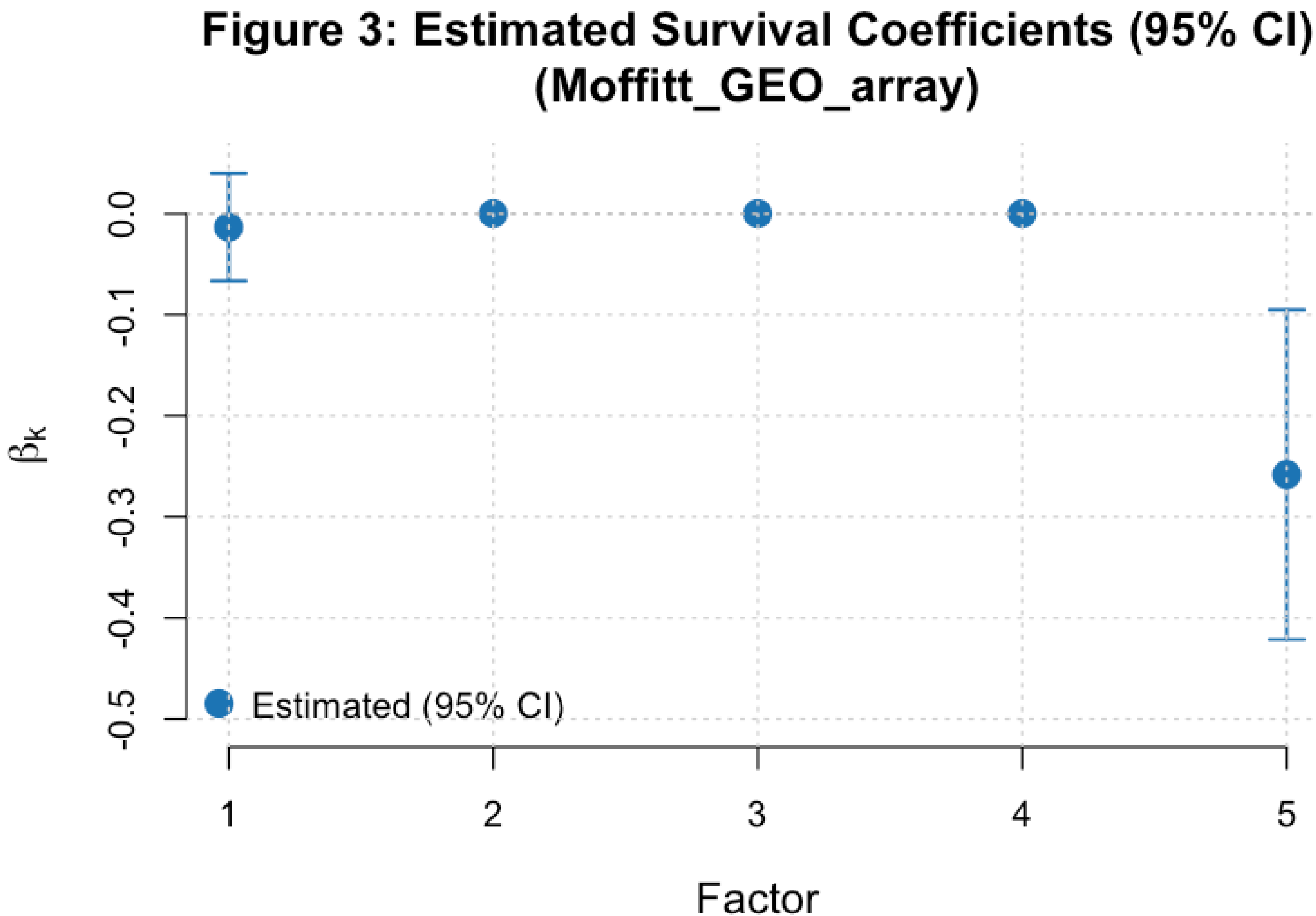

Figure 19: Moffitt

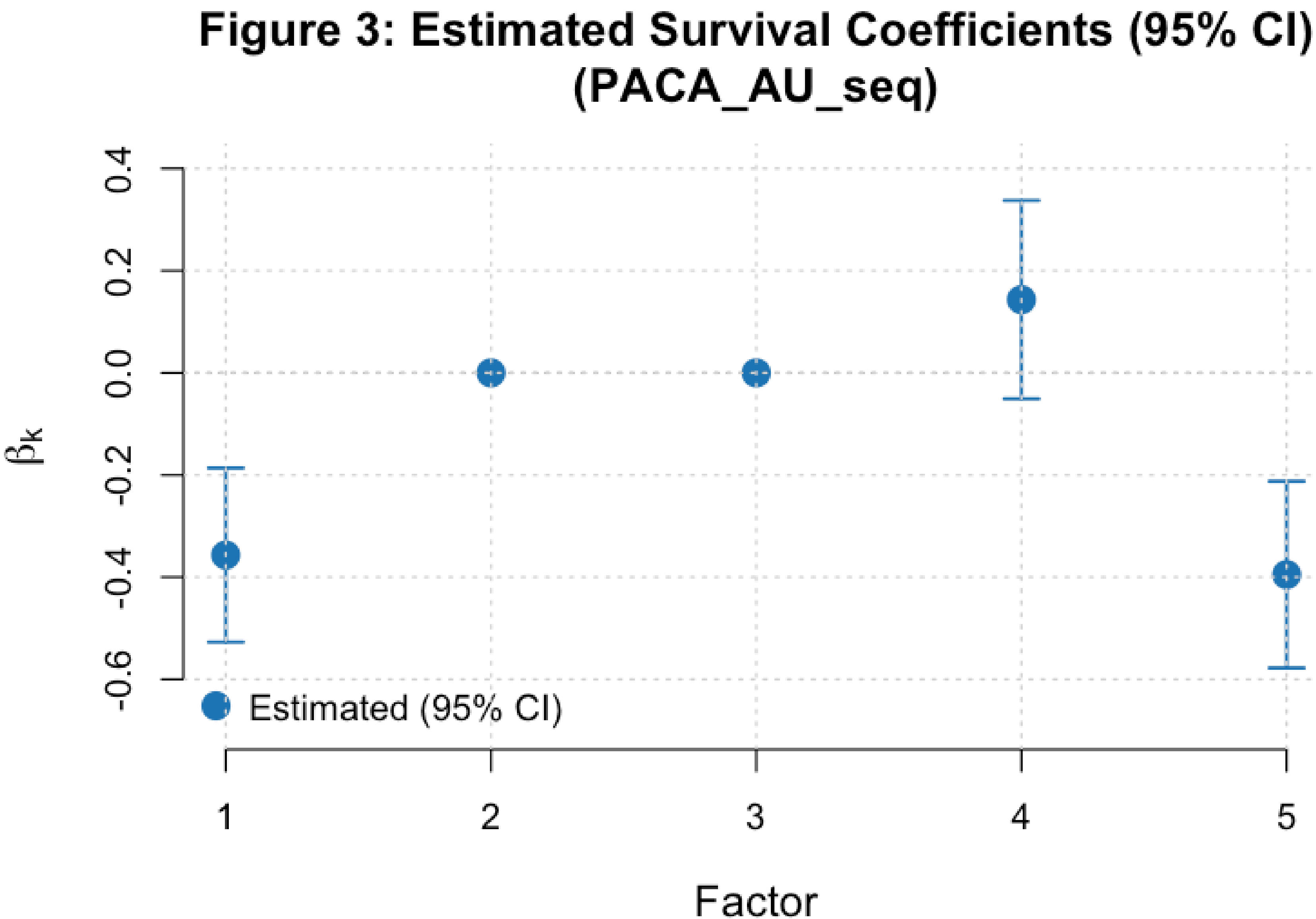

Figure 20: PACA_AU_seq

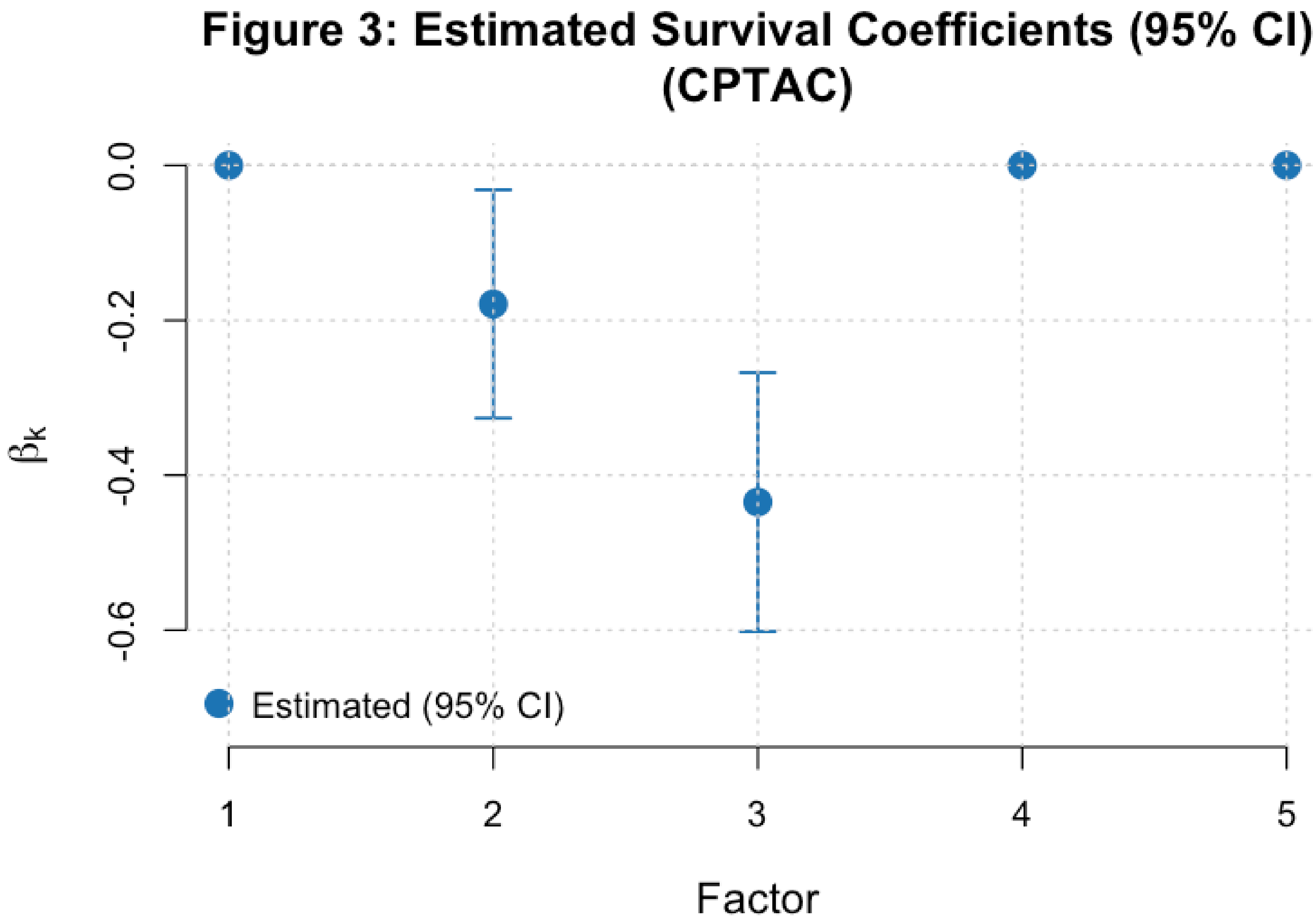

Figure 21: CPTAC

Per-Cohort Kaplan-Meier Curves

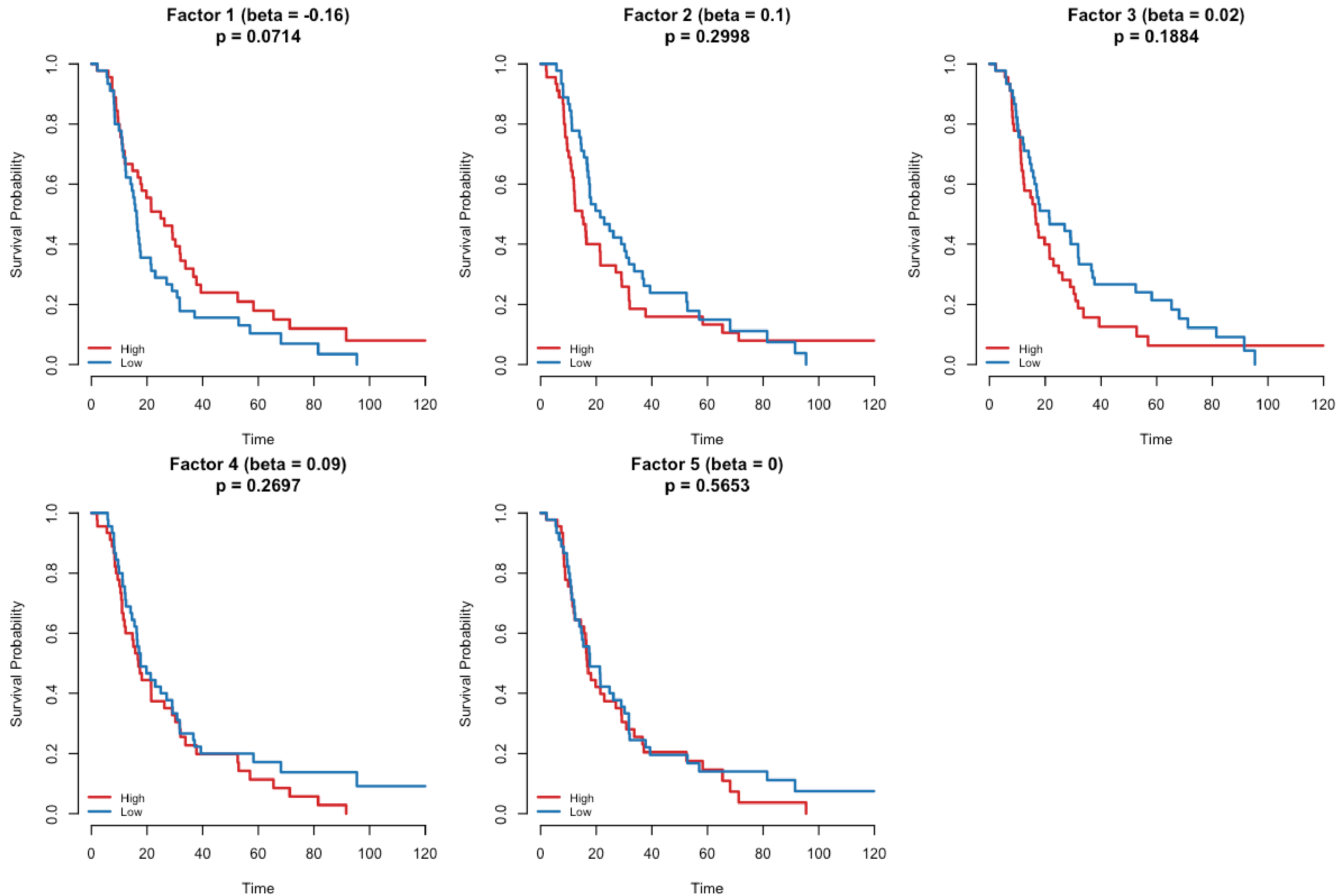

Figure 22: Dijk

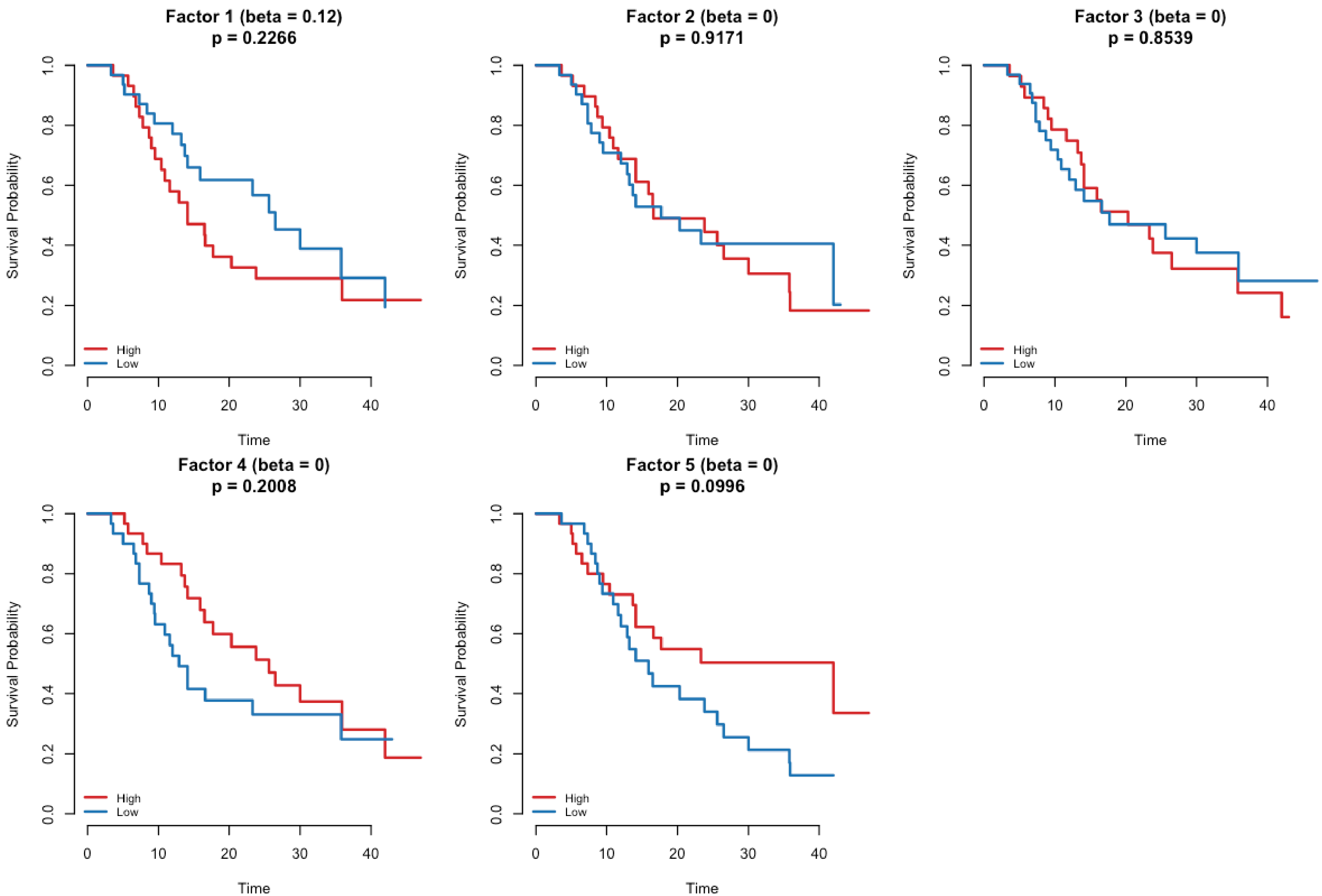

Figure 23: PACA_AU_array

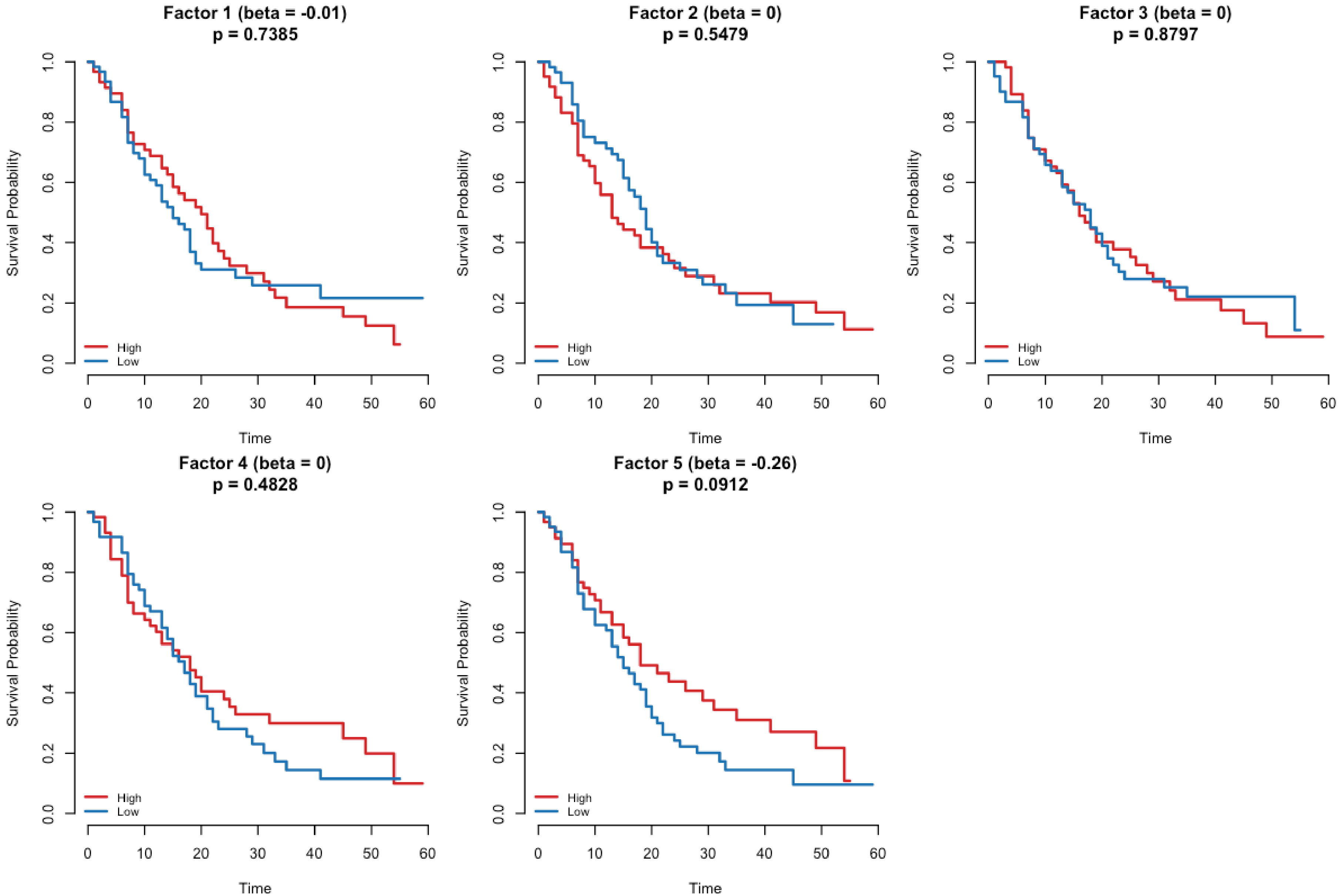

Figure 24: Moffitt

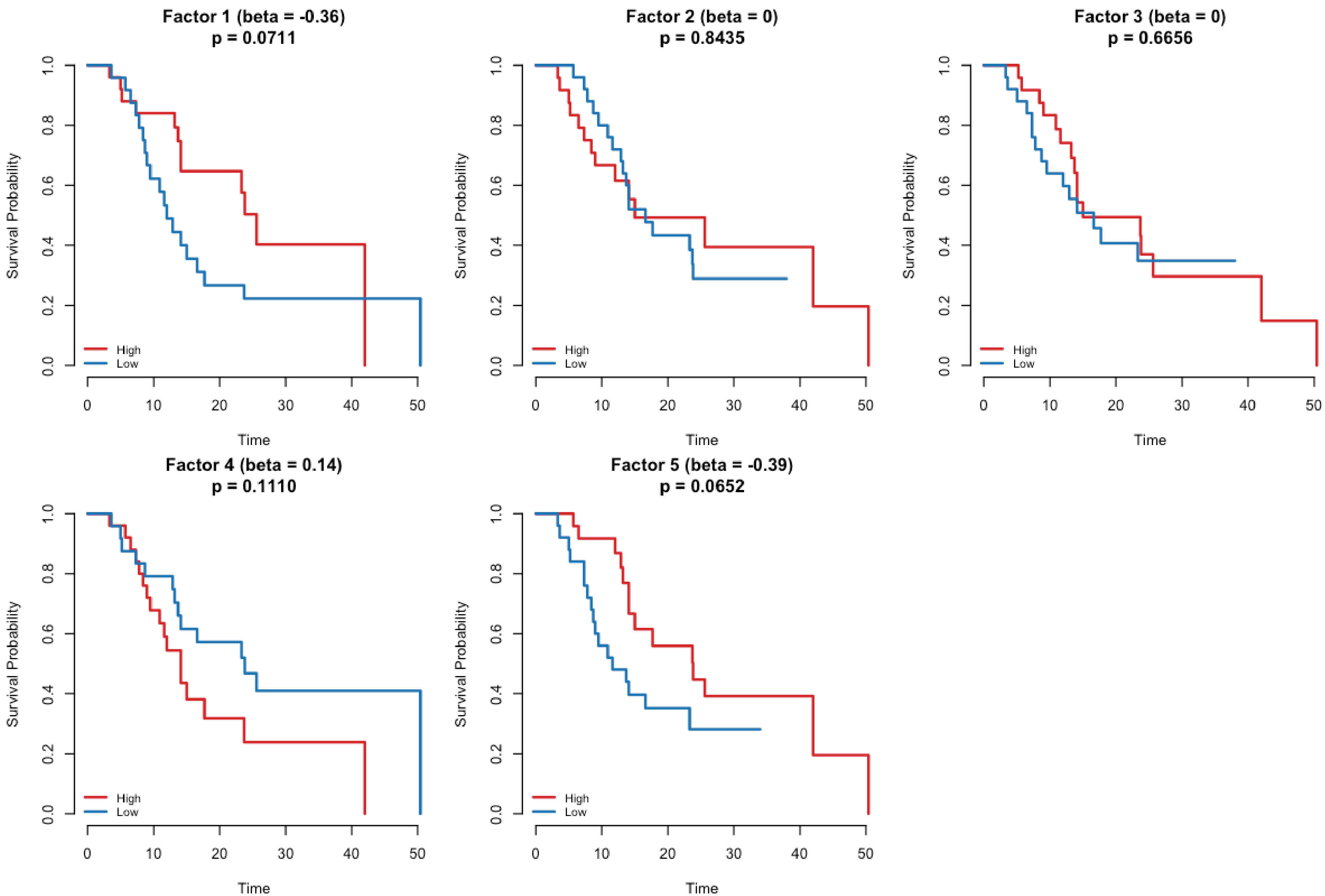

Figure 25: PACA_AU_seq

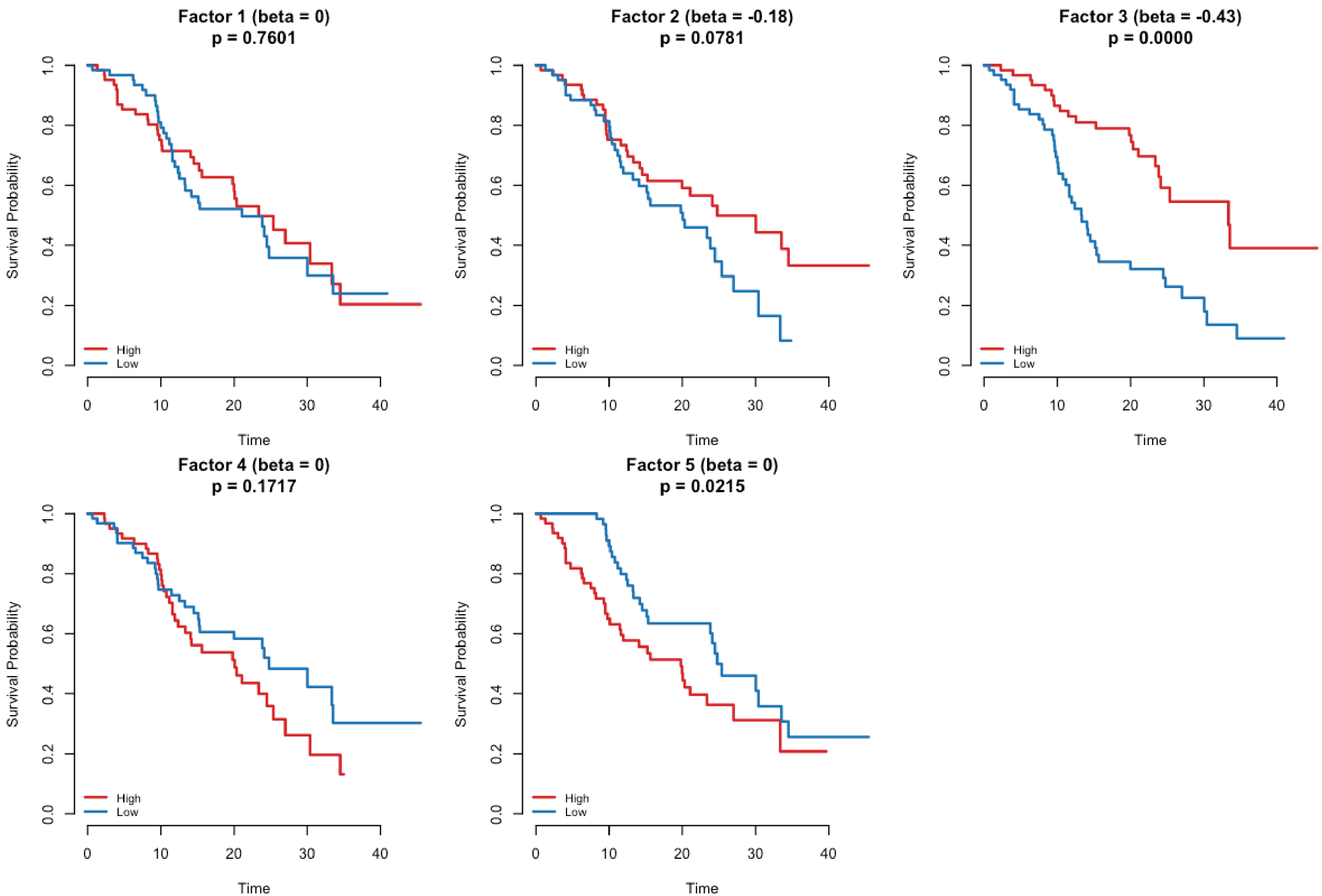

Figure 26: CPTAC

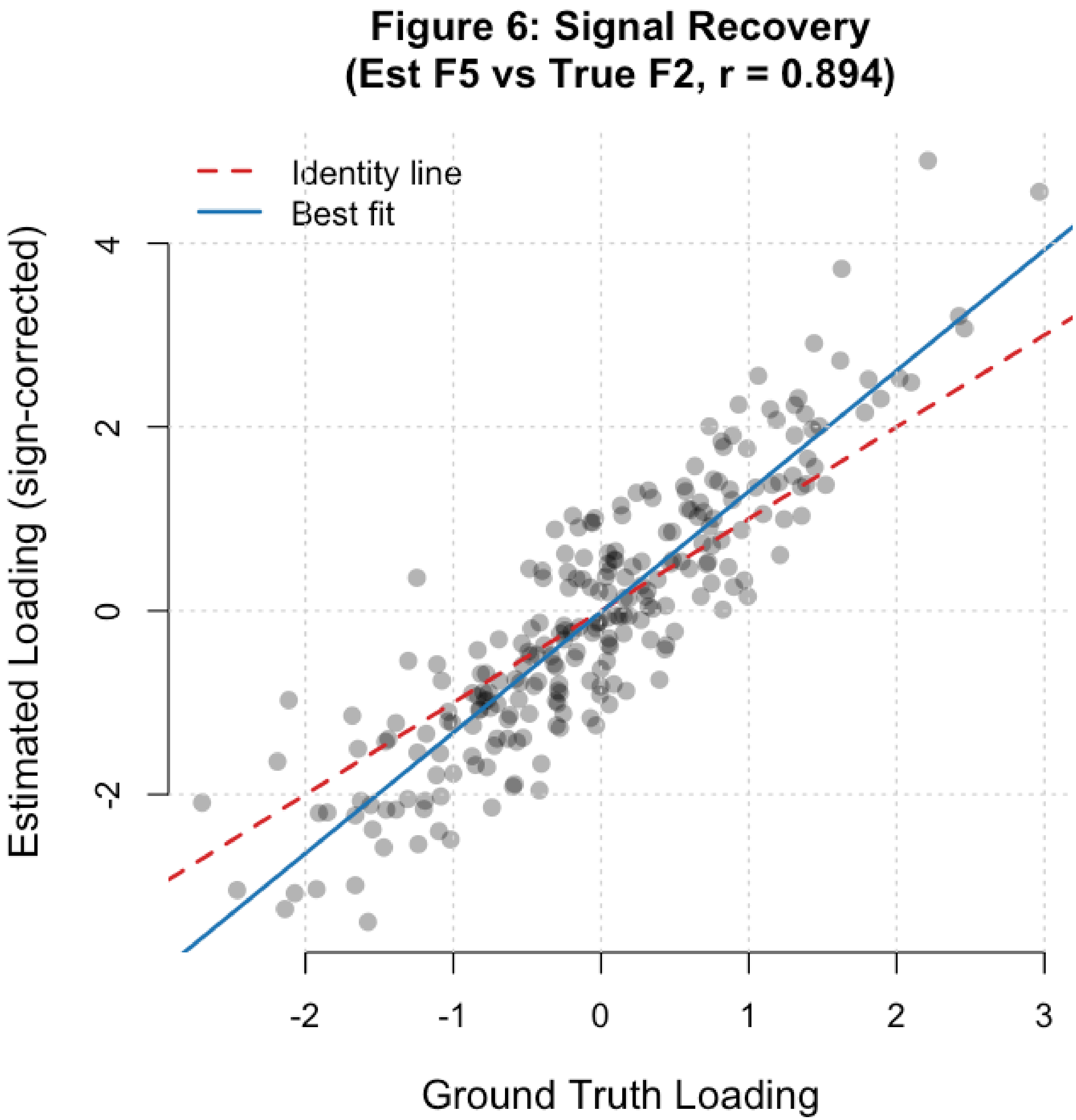
Figure 6: Signal Recovery
(Est F5 vs True F2, r = 0.894)
Identity line
Best fit
Estimated Loading (sign-corrected)
4
2
0
-2
-2
-1
0
1
2
3
Ground Truth Loading

Method Comparison

SBMF vs. Penalised Supervised MF Approaches

Feature	Penalised MF (e.g., deSurv)	SBMF (this work)
Inference	Penalised optimisation	Full Bayesian (CAVI)
Objective balance	Tuning parameter α	ELBO — natural balance
Priors	Fixed penalty λ	EBNM — prior learns from data
Uncertainty	Point estimates only	Posterior moments $(E[\theta], \mathrm{Var}(\theta))$
K selection	CV grid search	Auto-prune (K_{eff})
Constraints	Nonnegative (NMF)	Unconstrained (allows negative loadings)
Error-in-variables	Not addressed	$E[L^2] = \mathrm{Var}(L) + E[L]^2$ correction
Second moments	N/A	Prevents overfitting to uncertain loadings